

Three layers of realisation: hierarchy recovery, holonomy, information and contextuality

Abstract

This paper consolidates three bodies of work into one account. Each result carries its source and its status: proved, computed, measured, open, or closed by theorem.

1. *Hierarchy recovery* [H]. A gradient-free estimator (spectral initialisation, soft hand-off, inside-outside EM and split-merge) recovers a depth-4 probabilistic grammar to 0.001 nats per sequence of the true model in 131 s. Twelve transformer configurations never represent the level-3 latent symbol.
2. *Cohomological strand* [O]. Holonomy of known operations gives a genuine obstruction $[g] \in H^1(\Gamma, A)$. Marginal obstructions within one data source vanish (T0). The information site and the contextuality site are separated by a theorem.
3. *Results of the present work* [S]:
 - an exact identifiability criterion for holonomy estimated from data, with a calibrated test and its sample cost;
 - a direct construction from operations to outcomes, $[g] \neq 0 \iff \gamma \neq 0$, which is invisible to the probability layer;
 - for lossy operations: a closed form $\text{CF} = \max(0, 1 - |L|\varepsilon/2)$ on a cycle, and a holonomy-girth lower bound on general graphs;
 - the loop spectrum as a monoid-valued holonomy invariant;
 - a decorated Bregman–Čech nerve that carries orientation and phase, with a codimension criterion for when it can;
 - restriction and extension maps between the information and contextuality views;
 - a gated contextuality test for streams with *not-applicable* entries, with a pinned maximum-likelihood null.

The organising statement is that the three layers (operations, probability and outcomes) respond to the same data through different invariants, and that no invariant of one layer determines another.

Provenance tags.

- [H]: *Recovering latent hierarchy without gradient descent* (hierarchy paper and handoff).
- [O]: *Holonomy obstructions for composed local operations* (holonomy paper).
- [S]: this work, implemented in the `amb.vigneaux` package. It has 107 tests, and every numerical claim tagged [S] is produced by that code.

Status tags: proved, computed (exact computation or verified identity), measured (empirical, no proof), open, closed (settled by theorem, not to be reopened).

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1 The standing constraint

Theorem 1.1 (T0: single-source global section). *[O, Thm 7] proved Let $\{P_C\}_{C \in \mathcal{M}}$ be context marginals estimated from one sample $\mathcal{D}$. The empirical joint $P_{\mathcal{D}}$ restricts to each P_C , so the family extends. Consequently the Čech obstruction $\gamma(s)$ vanishes for every section and the contextual fraction is 0, on any cover, acyclic or not.*

Proof. $P_{\mathcal{D}}$ is a global section by construction. □

Corollary 1.2. *[H/O] proved A single fitted model can never exhibit an obstruction to its own consistency. If an estimator and a detector share a data source, the detector measures model misfit, not structure.*

Remark 1.3 (T0 as a functor). [S] proved Restriction R from global laws to context families (Section 23) has image exactly the non-contextual models. So T0 is the statement that anything in the image of R has $\text{CF} = 0$ and $\gamma = 0$. The engine’s first version violated T0 by resampling a single-source world context by context: it produced a spurious CF up to 0.10, and the fitted model’s CF equalled its signalling defect exactly. After adding a single-source sampling mode, $\text{CF}(\hat{P}_N) \leq 2 \times 10^{-16}$.

2 The three layers

Layer	Invariant	Condition for non-triviality
1 Operations	$H^1(\Gamma, A) \cong A^{\beta_1}$, holonomy; the loop spectrum for lossy operations	The operations are known or identifiable (Prop. 15.1); $\beta_1 > 0$
2 Probability	$H^1(\mathcal{S}; \mathcal{F}_\alpha)$, one-dimensional, generated by entropy (Tsallis for the α -action)	The conditioning action is present; $I = \delta_t H$ measures its failure to be trivial
3 Outcomes	$\gamma \in \check{H}^1(\mathcal{U}, \mathcal{F}_{\tilde{\mathcal{C}}})$; CF	The cover has no global context; data come from more than one source (T0)

Table 1: The stratification [O, §6], extended [S].

Remark 2.1 (Stratification, not bridge). [O, Rem. 18] proved Each invariant is non-trivial on its own layer and zero on the others, and the reasons are the hypotheses themselves. Layer 2’s site has a global joint, which gives the observable monoid an absorbing element; Theorem 13.3 then kills every trivially-acting coefficient system, which is where Layer 3’s coefficients would have to live. Layer 3’s site is built so that no global context exists. That is what permits $\gamma \neq 0$, and it simultaneously denies Layer 2 the joint it requires.

[S] adds that the layers are connected *around* Layer 2, not through it. There is an exact construction from Layer 1 to Layer 3 (Theorem 16.1), and Layer 2 is provably blind to it (Proposition 16.2).

Part I

Hierarchy recovery [H]

3 Setup and exact floors

All experiments use the Random Hierarchy Model of Cagnetta et al. [9]. It is a probabilistic context-free grammar on a fixed complete binary tree of depth L , with V symbols per internal level, m productions per symbol, and a leaf alphabet of size A . Because the tree shape is fixed and positional, the latent structure is a Markov chain from leaves to level 1 and on up to the root.

- *Corpus A*: $L = 4$, $V = 64$, $A = 48$, $m = 12$, 2×10^5 sequences of length 16.
- *Corpus B*: $L = 4$, $V = 8$, $A = 12$, $m = 3$, 2×10^5 sequences.

The exact conditional entropy $H(x_t \mid x_{<t})$ is computed by inside–outside (Theorem 10.1). For Corpus A: $H_{\text{local}} = 1.6034$, $H_{\text{higher}} = 3.5527$, $H(x_0) = 3.8161$ nats. The latent-symbol entropies given the prefix, for levels 1–4, are 2.7835, 2.2923, 2.2900, 2.7910 against chance $\ln V = 4.1589$. About 1.87 nats of level-3 information are recoverable from the prefix.

4 Transformers do not recover levels 3–4

configuration	local	higher	level-3 probe
2 layers, $d = 64$ (baseline)	2.173	3.831	0.028
4 layers	2.078	3.797	—
8 layers, frozen embedding	2.037	3.755	—
8 layers, frozen emb., fresh sampling	2.022	3.772	0.027
$d = 128$, 4 layers	2.039	3.770	—
frozen random embedding (2 layers)	2.186	3.824	—
constructed order coordinate	2.162	3.815	—
auxiliary supervision, levels 3–4	2.088	3.777	0.056
latent marginalising head (stopped at 34k)	3.227	3.835	—
dilated causal CNN	2.104	3.827	0.032
curriculum over tree depth	2.085	3.803	0.034
curriculum control (no easy cases)	2.043	3.789	0.033
exact floor	1.603	3.553	—

Table 2: Corpus A, 48,000 steps each. Shuffled-label controls read 0.021–0.034; chance is $1/V = 0.0156$. [H]

Statement [H] *measured*. Across twelve configurations the level-3 probe never leaves the shuffled-label floor, although 1.87 nats of level-3 information are present in the prefix. A linear probe on the embedding was *withdrawn*: a random matrix supports the same probe accuracy.

Hypothesis (not established). Recovering the level- ℓ symbol requires marginalising over the parses consistent with the prefix. A feedforward stack has only an upward pass, while inside–outside couples an upward and a downward pass.

5 The estimator and its verification

Construction 5.1 (Estimator). [H]

1. *Spectral initialisation*: build the pair-context co-occurrence matrix, take its leading singular directions, and k -means the pairs into V groups.
2. *Soft hand-off*: pass the posterior to the next level, $q_\ell \propto \sum_{x,y} P_\ell[\cdot, x, y] q_{\ell-1}^{\text{lo}}(x) q_{\ell-1}^{\text{hi}}(y)$.
3. *EM*: inside-outside E-step and normalised-counts M-step, keeping the sweep with the best held-out likelihood.

check	statement	max error
V0	EM never decreases the training log-likelihood	$+1.8 \times 10^{-2}$ (min step)
V1a	a free subtree has inside value exactly 1	1.8×10^{-15}
V1b	restriction = marginalisation	8.9×10^{-16}
V1c	gluing = independence join, $P(x_{0:16}) = \sum_{u,v} \mu(u, v) \beta_L(u) \beta_R(v)$	1.0×10^{-14}
V2	the soft hand-off is a sufficient statistic	6.7×10^{-16}
V4a	merge cost = $N I(\text{children}; B \mid q(B))$	1.2×10^{-14}

Table 3: Identities of the tree model class [H] computed. They verify the implementation and that the fitted object is a sheaf section. They do not test whether the data are tree-generated. The hard (argmax) decode flips on 34–46% of sequences under a chain of conditionings; the soft decode does not change.

variant (Corpus A, 2×10^4 sequences)	local	higher	held-out log p /seq
joint EM from random init	2.566	3.831	−51.18
bottom-up, frequency-rank clusters	2.598	3.833	−51.45
spectral + soft hand-off + EM (10 sweeps)	2.227	3.802	−48.23
spectral + soft hand-off + EM (50 sweeps)	2.162	3.807	−47.75
stagewise + vertical re-clustering	2.240	3.802	−48.34
transformer, 48k steps	2.022	3.772	—
true grammar	1.599	3.553	−41.21

Table 4: Estimator variants on Corpus A [H].

6 Instruments for detecting hierarchy

- **V6, leaf pairs** [H] measured. The test is a parametric bootstrap of empirical mutual information against datasets sampled from the fitted model.
 - *Planted positive*: copying leaf 7 into leaf 8 with probability 0.3 gives $z = +220$, against +6.5 elsewhere; even the true grammar is refuted there ($z = +390$).
 - *Limitation*: on Corpus A, pairwise leaf MI across level-2 and root boundaries is about 10^{-4} even under the true grammar, while the plug-in bias floor is about 0.055. So V6 has no power above level 1.
- **V7, blocks** [H] measured. Each block is decoded by its own inside vector, and adjacent blocks are compared under the disjointness invariant.

- Decoded with the true grammar, blocks sharing a parent carry about 1 nat, and others 0.002–0.014.
- The fitted model captures 79%, 1.4% and 3.9% of this at levels 1–3.
- β_1 of the block complex [H] measured. A tree predicts $\beta_1 = 0$ at every threshold.
 - Corpus B, unplanted: 0 at all 25 thresholds.
 - Two planted block couplings: $\beta_1 = 1$ at 5 thresholds, with the null at 0. A single coupling never produces a cycle.
 - β_1 is computed on a Rips complex, so it is a calibrated statistic and not an approximation to Čech persistence (Theorem 10.20).

7 The regime determines learnability

	level 1	level 2	level 3	level 4
Corpus A: observations per parameter	1.4	0.31	0.15	0.08
Corpus B: observations per parameter	1389	1563	781	391

Table 5: Identifiability regime [H]. Levels fail at 0.15 observations per parameter and succeed at 781.

On Corpus B at 200 sweeps, the level-3 block dependence captured rises from 0.0854 to 0.1827, and its discrepancy falls from $z = +20.2$ to $+5.4$. The earlier failure was therefore an unestimable regime, then undertraining. Merges at levels 1–3 cost real conditional MI; the root alphabet is redundant, with $8 \rightarrow 2$ improving held-out likelihood by 0.004.

8 Closing the gap: basin selection and split–merge

start	held-out log p /seq	local / higher
exact truth ($\varepsilon = 0$)	−15.324	0.3322 / 1.5833
truth + 50% / 80% / 90% uniform	−15.325 / −15.325 / −15.330	—
truth + 95% / 99% uniform	−15.775 / −18.113	—
uniform	−20.706	0.5091 / 2.0791
spectral init (ours)	−17.511	0.3952 / 1.7937

Table 6: Diagnosis: EM started at the truth [H] measured. The truth is a fixed point with a very wide basin.

A hard partition of observed child pairs cannot represent a grammar in which pairs are shared across parents (4 of 16 at level 1).

- Sharpening the clustering hurts: sibling context gives −18.06.
- Blurring it helps: blending with uniform at weight 0.9 gives −15.87.

Construction 8.1 (Split–merge). [H] For each level:

1. duplicate the symbols with a perturbation;

2. run EM;
3. greedily merge back by Bregman information over *all* $2V$ symbols, so merges can recombine across the split;
4. accept only if held-out likelihood improves.

stage	held-out $\log p/\text{seq}$	merge cost (nats/seq), levels 1–4
warmup, 40 EM sweeps	−17.187	—
round 1	−15.491	0.160, 0.113, 0.008, 0.003
round 2	−15.325	0.042, 0.011, 0.007, 0.005
round 3	−15.3253	0.052, 0.020, 0.007, 0.004
true grammar	−15.3243	—

Table 7: Split–merge on Corpus B [H] *measured*: 131s, local/higher 0.3322/1.5835 against a true 0.3322/1.5833.

The recovered model passes all four instruments:

- V3 matches the true column at every boundary.
- V6 flags 0 of 18 pairs, with largest $|z| = 1.7$.
- V7 at level 3 reads 0.5863/0.5882 against a true 0.6140.
- β_1 against the fitted null is 0 of 25, where the local-optimum fit gave 6 of 25.

9 Why clustering is necessary at scale

A dense inside pass costs about $2NV^3$. At $N = 10^6$ and $V = 1024$ a sweep costs about 4.3×10^{15} . Compression to $K = 16$ gives about 8×10^{11} over 50 sweeps, a speedup of $(V/K)^3 = 262,144$. For m -ary grammars the order rises to $m + 1$, so compression buys $\log_K V$ orders of headroom.

Compression is not free. Merge costs are level-dependent and must be measured, not assumed; V4a supplies the measurement. At $L = 16$ the root is observed only about 15 times per 10^6 tokens, so compression removes the compute barrier but not the identifiability barrier.

10 Theorems and constructions used [H]

Theorem 10.1 (Inside–outside; Baker, Lari–Young). *used* For a PCFG on a fixed tree, the inside values β and outside values α give exact marginals: the posterior at a node is $\propto \alpha\beta$.

Theorem 10.2 (EM ascent; Dempster–Laird–Rubin). *used* Each EM iteration does not decrease the observed-data log-likelihood.

Theorem 10.3 (Convergence; Wu). *used* Under regularity conditions, EM converges to a stationary point, and the limit depends on the initialisation.

Theorem 10.4 (Extension; Vorob’ev). *used* Every consistent family of marginals on a hypergraph extends to a global distribution if and only if the hypergraph is acyclic. A fixed tree is acyclic.

Theorem 10.5 (Independent pullback; Simpson Thm 6.4, Prop. 8.5). *used* A presheaf on standard Borel probability spaces is an atomic sheaf iff it maps independent squares to pullbacks. The independent pullback is the conditional product.

Definition 10.6 (Supports; Simpson Def. 7.1). A support of a random variable is its terminal factorisation through a sample space.

Theorem 10.7 (Conditional products; Fritz §12). *used* In a Markov category with conditionals, conditional products exist and satisfy the Dawid–Studený axioms. FinStoch is such a category.

Theorem 10.8 (Sufficient statistics; Fritz Def. 14.3, Thm 14.5). *used* A statistic is a deterministic morphism, and the abstract Fisher–Neyman factorisation holds. The soft hand-off is a statistic, and V_2 verifies its sufficiency.

Theorem 10.9 (Minimal sufficiency; Fritz Def. 16.2, Bahadur). *used* A minimal sufficient statistic is least informative among the sufficient ones; a complete sufficient statistic is minimal.

Remark 10.10 (Fritz and cohomology). Fritz’s framework supplies semantics for conditional independence and sufficiency, but no cohomology. It does not bear on a map from information cohomology into a persistence module.

Construction 10.11 (Independence join; Wright §7.1). For tables agreeing on a shared marginal, $(p \bowtie q)_{ijk} = p_{ij}q_{jk}/r_j$ is the maximum-entropy weak pullback.

Construction 10.12 (Merging as convex combination; Wright §7.2). Merged rows are mixtures, averaged in the Giry monad.

Theorem 10.13 (Bregman information; Bregman, Banerjee et al.). *used* For KL , the optimal cluster representative is the mean, and the clustering loss equals the mutual information between the data and the cluster assignment.

Construction 10.14 (Merge criterion). Merging symbols costs

$$\sum_s n_s D_{\text{KL}}(R_s \| R_{\text{class}}) = N I(\text{children}; B \mid q(B)).$$

The cost is zero iff the child rows are equal, which is strong (downward) lumpability (Kemeny–Snell). Observational losslessness is strictly broader: upward-equal merges lose nothing at positive cost (Proposition 25.5).

Construction 10.15 (Agglomerative merging; Slonim–Tishby). Greedy least-loss merging is the agglomerative information bottleneck; split–merge (Petrov et al.) is its grammar analogue.

Theorem 10.16 (Ostrowski). *used* Every nontrivial absolute value on $\mathbb{Q}$ is equivalent to the real absolute value or a p -adic one, and the resulting completions are pairwise inequivalent.

Construction 10.17 (Bias correction; Miller–Madow). Plug-in MI on a $K \times L$ table is biased upward by about $(K-1)(L-1)/2N$.

Construction 10.18 (β_1). $\beta_1 = \dim \ker \partial_1 - \text{rank } \partial_2$, computed over $\text{GF}(2)$ on the 2-skeleton.

Proposition 10.19 (The lattice quasi-morphism does not exist). *[H, Prop. 1] proved, closed* Let $\rho : \mathbb{Q} \rightarrow L$ be order-preserving for the real order and constant on each p -adic ball. Then ρ is constant.

Proof. A p -adic ball is dense in $\mathbb{R}$. Choose $x < y < z$ with x, z in one ball B_1 and y in another ball B_2 . Monotonicity gives $c_1 \leq c_2 \leq c_1$. $\square$

Theorem 10.20 (Bregman TDA; Edelsbrunner–Wagner). *used KL restricted to the simplex is of Legendre type. Čech and Delaunay complexes of Bregman balls have the homotopy type of the union (Contractibility Lemma). The Rips filtration can be arbitrarily far from Čech (Result 2). Measured [H]: maximum β_1 is 20–21 for Čech and 0 for Rips.*

Refuted, do not retry [H]:

- sibling-only clustering context (−18.06);
- deterministic annealing (+0.22 nats, independent of T_0);
- stagewise plus vertical re-clustering (−48.34 against −47.75);
- posterior entropy as a confidence criterion: posterior entropies are 0.198–1.003 at levels capturing under 4%, whereas $I(\text{parent}; \text{children})$ reads 10.1, 8.1, 4.5, 5.3 against a true 8.9–9.0;
- curriculum transfer: the control gives 2.043/3.789 against 2.085/3.803.

Part II

Holonomy and the separation of sites [0]

11 The holonomy obstruction

Definition 11.1 (Operation system). [O, Def. 1] (Γ, A, g) consists of a finite connected graph $\Gamma = (V, E)$, an abelian group A , and $g : E \rightarrow A$ with $g_{ba} = -g_{ab}$.

Definition 11.2 (Global field). [O, Def. 2] A map $c : V \rightarrow A$ with $c(b) - c(a) = g_{ab}$ for every edge.

Theorem 11.3 (Holonomy obstruction). [O, Thm 3] *proved, computed* A global field exists iff the holonomy $h(L) = \sum_{e \in L} \pm g_e$ vanishes on every cycle. The obstruction is $[g] \in H^1(\Gamma, A) \cong A^{\beta_1}$, with $\beta_1 = |E| - |V| + 1$.

Proof. Integrate g along a spanning tree. A non-tree edge is consistent iff its fundamental cycle has zero holonomy, and the fundamental cycles form a basis of the cycle space. $\square$

Verified against brute force on a room with $|V| = 4$, $|E| = 5$, $\beta_1 = 2$, $A = \mathbb{Z}/4096$. The holonomy is $(0, 0)$ when a field exists, and $(0, 1)$ and $(2, 1)$ when none does [O]. The same computation in [S] gives the same classes, up to a unimodular change of cycle basis.

Theorem 11.4 (Holonomy as a connecting class). [O, Thm 4] *proved* Consider the exponential sequence $0 \rightarrow \mathbb{Z}_M \rightarrow \mathbb{R}_M \rightarrow \mathcal{S}_M^1 \rightarrow 0$ on a manifold M with boundary. Since $\mathbb{R}_M$ is soft, the connecting map $\delta_M : H^0(M, \mathcal{S}^1) \rightarrow H^1(M, \mathbb{Z})$ is an isomorphism modulo global real lifts. Naturality under $r : \partial M \hookrightarrow M$ gives $r^*(\delta_M[g]) = \delta_{\partial M}(r^*[g]) = \text{Hol}(\alpha, \partial M)$.

Remark 11.5 (What Theorem 11.4 does not say). [O, Rem. 5] $\delta_M[g]$ lies in $H^1(M, \mathbb{Z})$, not in $\text{Ext}^1(\mathcal{S}_M^1, \mathbb{Z}_M)$. It is not the AMB obstruction: the “support reduction” $\mathcal{F}_{\text{prob}} \rightarrow \mathbb{Z}$ is not additive, so it is not a morphism of sheaves.

Proposition 11.6 (T0 does not apply to operation systems). [O, Prop. 8] *An operation system has no sample; a witness would be a global field, which is the object in question.*

Proposition 11.7 (Vorob’ev does not apply). [O, Prop. 9] *Here the relevant structure is the cycle space of Γ : $\beta_1 = 0$ makes H^1 trivial, while $\beta_1 > 0$ leaves it free.*

Proposition 11.8 (Possibilistic fragility does not apply). *[O, Prop. 10] computed The AMB obstruction is a functional of the support, so smoothing kills it. Measured: γ falls from 9/9 to 0/27 at $\varepsilon = 10^{-9}$, while $\text{CF} = 1 - 3\varepsilon$ survives to $\varepsilon = 1/3$. The holonomy class is a sum of the operations themselves and has no support to blur.*

attempted coefficients	why it failed	verified
p -adic lattice $\rho : \mathbb{Q} \rightarrow L$	order-preserving and constant on balls implies constant	proof
probability simplex Δ^{V-1}	a convex set with no inverses, so no δ	—
support indicator $\omega = \int_{\text{supp}} \mathbf{1} dP$	identically 1 on every face of every family	computed
rounding $\mathbb{R} \rightarrow \mathbb{Z}$	not additive, so not a module map	—
$\mathbb{Z}/n$ on edges	abelian, δ exists, the class varies	computed

Table 8: The coefficient group is what mattered [O, §3].

Remark 11.9 (β_1 changes role). [O, Rem. 11] In the marginal setting, β_1 of the incidence complex is a function of the cover alone. It equals 4 for the obstructed 3×3 parity family and for both controls, so it cannot map to γ . In an operation system, β_1 is the rank of the obstruction group.

12 The two coboundary operators

Definition 12.1 (Conditioning action). [O, Def. 12] $(S_0 \cdot F)(P) = \sum_j P(S_0 = v_j) F(P \mid S_0 = v_j)$.

Definition 12.2 (The two coboundaries). [O, Def. 13]

$$(\delta F)(S_0; \dots; S_k) = S_0 \cdot F(S_1; \dots; S_k) + \sum_{i=1}^k (-1)^i F(\dots; S_{i-1} S_i; \dots) + (-1)^{k+1} F(S_0; \dots; S_{k-1}).$$

δ_t is the same expression with the first term replaced by $F(S_1; \dots; S_k)$. Both operators square to zero.

Theorem 12.3 (Baudot–Bennequin). [O, Thm 14]; [5] used; verified numerically [S]

1. $\delta c = 0$, so $H^0 \cong \mathbb{R}$.
2. $\delta H = 0$ (the chain rule), and under locality hypotheses $H^1(\mathcal{F}, \delta)$ is one-dimensional, generated by H .
3. $\delta_t H = I$, and $(\delta - \delta_t)H = (S_0 - 1) \cdot H(S_1) = -I$.

[S] verifies numerically: $\delta H = 0$ and $\delta_t H = I$ to 10^{-10} ; $(XY) \cdot F = X \cdot (Y \cdot F)$; $\delta^2 = 0$; and Rényi-2 entropy is not a cocycle.

Remark 12.4 (Easy to get backwards). [O, Rem. 15] $I_2 = \delta_t H$, not δH . Higher I_k are coboundaries for δ or for δ_t according as k is odd or even. No regularity or symmetry condition separates H from I .

13 Non-unification and vanishing

Theorem 13.1 (Non-unification). [O, Thm 16] proved No morphism of coefficient systems carries a nonzero AMB class in $\mathcal{F}_{\mathbb{Z}}$ to a nonzero class in $\mathcal{F}_{\alpha}$.

Remark 13.2 (Holonomy is not the bridge). [O, Rem. 17] The holonomy class shares its coefficient type with AMB and its locality with Vigneaux, but it is a cocycle of transformations, not of sections or information functions. So it is a third object.

Theorem 13.3 (Vanishing on the trivially-acting complex). [O, Thm 27] *proved, computed* Let $\mathcal{S}$ be an information structure whose observables form a monoid under joint, with an absorbing element, and let A be an abelian group with trivial action. Then $H^n(\mathcal{S}; A) = 0$ for $n > 0$ and $H^0 = A$.

Proof. With trivial action the cochains are constants, and the complex is the bar complex of a monoid with an absorbing element, whose classifying space is contractible. $\square$

Verified for the join-semilattices of two and three binary variables: $H^0 = \mathbb{Z}$ and $H^1 = H^2 = 0$ over $\mathbb{Q}$, and over $\mathbb{Z}$ by Smith normal form, with no torsion.

Corollary 13.4 (Extensions do not help). [O, Cor. 28] *proved, closed* The connecting map of any extension of $\mathcal{F}_\alpha$ by a trivially-acting $\mathcal{F}_\mathbb{Z}$ lands in $H^{k+1}(\mathcal{F}_\mathbb{Z}) = 0$.

Remark 13.5 (The dichotomy). [O, Rem. 29] Either the coefficients act trivially, and Theorem 13.3 applies; or they do not, and the AMB class does not live there. An information structure presupposes that every set of variables has a joint. A measurement scenario presupposes that some do not.

Remark 13.6 (Scope of Theorem 13.3). [S] The theorem concerns the *trivially-acting* complex. Under the non-trivial conditioning action, H^1 is one-dimensional (Theorem 12.3); under the α -deformed action it is generated by Tsallis entropy (Section 18). $H^{\geq 2}$ is open for both actions (Baudot–Bennequin, Problem 1). The finite linear algebra of `h2_rational.py` does *not* transfer to the deformed action: cochains are then functions of P , and the spaces are infinite-dimensional without a restricted function class.

Problem 13.7 (Site correspondence). [O, Prob. 21] *open* Does an AMB scenario determine an information structure in Vigneaux’s sense?

Problem 13.8 (Cover preservation). [O, Prob. 22] *open* Is the correspondence a morphism of sites? This depends on Problem 21.

Problem 13.9 (Coefficient comparison). [O, Prob. 23] *blocked* A morphism $\Psi : \mathcal{F}_\mathbb{Z} \rightarrow \mathcal{F}_\alpha$ intertwining the module structures has an undefined left-hand side.

Problem 13.10 (Extension formulation). [O, Prob. 24] *settled negatively* by Cor. 28

Problem 13.11 (Parity criterion). [O, Prob. 25] A mod- p family is contradictory iff every variable lies in an even number of contexts and $\sum_C s_C \not\equiv 0 \pmod{p}$. Brute force caught two failed examples. [S] reproduces the mod-3 grid exactly (Section 14).

Problem 13.12 (Holonomy from estimated operations). [O, Prob. 26] *answered in part* by §15

Closed by theorem, not to be reopened [O]:

- the p -adic bridge (Prop. 10.19, Theorem 10.16);
- branch cohomology with simplex coefficients: a rooted tree poset has an extremal object, so $H^{k>0} = 0$ for every coefficient system;
- a map from β_1 of the incidence complex to γ ;

- a class for $I(X; Y \mid Z)$ obtained by regularity restrictions;
- the support-reduction identification of boundary holonomy with γ .

Part III

Results of the present work [S]

14 Engine and verification of the outcome layer

The `amb.vigneaux` package implements three things:

- the outcome layer: support presheaves, logical and strong contextuality, and γ decided exactly over $\mathbb{Z}$ (by integer diagonalisation), $\mathbb{Q}$ and $\mathbb{Z}/p$;
- the contextual fraction by LP, with its dual Bell inequality;
- the probability layer: conditioning action, δ , δ_t and co-information.

It reproduces the standard values *computed*:

- PR box and GHZ: strongly contextual, $\text{CF} = 1$, $\gamma \neq 0$ at every section.
- Tsirelson box: $\text{CF} = \sqrt{2} - 1$; noisy Tsirelson: $\text{CF} = \max(0, \sqrt{2}v - 1)$; noisy PR box: $\text{CF} = \max(0, 2v - 1)$.
- Hardy model: logically contextual with $\text{CF} = 1/6$, and a *false negative* of $\mathbb{Z}$ -cohomology. The returned witness family uses a -1 coefficient and is verified compatible, which agrees with the note in `glue.py`.
- Mod-3 grid (Problem 13.11): $\gamma \neq 0$ at 9/9 over $\mathbb{Z}$ and $\mathbb{Z}/3$; 0/9 over $\mathbb{Q}$, $\mathbb{Z}/2$ and $\mathbb{Z}/5$. Both controls give 0; $\text{CF} = 1, 0, 0$; the fragility of Proposition 11.8 is reproduced exactly.

15 Problem 26: holonomy from estimated operations

Operations are observed through a *design*: integer functionals $a_k \in \mathbb{Z}^E$, with observations $y_k = a_k \cdot g + \epsilon_k$ and a noise law independent of g .

Proposition 15.1 (Ring-theoretic identifiability). *proved, computed* Let $A \in \{\mathbb{R}, \mathbb{R}/\mathbb{Z}, \mathbb{Z}/n\}$ with $R = \mathbb{Q}, \mathbb{Z}, \mathbb{Z}/n$ respectively. The holonomy $h(L) = \ell_L \cdot g$ is identifiable iff $\ell_L \in \text{span}_R\{a_k\}$.

Proof. If $\ell_L = \sum \lambda_k a_k$, then $h = \sum \lambda_k (a_k \cdot g)$. Conversely, A is an injective cogenerator over R . If $\ell_L \notin \text{span}$, there is g' with $a_k \cdot g' = 0$ for all k and $\ell_L \cdot g' \neq 0$, so g and $g + g'$ are observationally equivalent. $\square$

Observing only each loop traversed twice identifies h over $\mathbb{Q}$ and over $\mathbb{Z}/\text{odd}$, but not over $\mathbb{R}/\mathbb{Z}$ or $\mathbb{Z}/\text{even}$. Witness pairs are constructed in the tests.

Corollary 15.2 (T0 for operations). *proved*

- A design that reads each vertex through one route (a spanning-tree section) spans no cycle, so no loop is identifiable.
- An estimator that fits a field and sets $\hat{g} = \delta \hat{c}$ returns $\hat{h} \equiv 0$. Measured: 10^{-17} on a world with $h = 0.2$, while edge-local estimates give 0.205.

Proposition 15.3 (Calibrated test and sample cost). *measured; large-sample* The Wald statistic $T = \hat{h}^\top \Sigma^{-1} \hat{h}$, with $\Sigma = C \text{diag}(\text{se}^2) C^\top$, is $\chi^2_{\beta_1}$ under $[g] = 0$ when the circular-mean standard error is the delta-method one, $\text{var} = (1 - \rho_2)/(2m\rho_1^2)$.

- Calibration: 4.2–6.4% false positives at nominal 5% for $\sigma \in [0.05, 0.35]$ turn ($m = 30$, 500 replicates). The naive standard error reaches 74%.
- Power follows the non-central χ^2 .
- Readings per edge for 80% power:

$$N_{80} = \frac{|L|}{h^2} v(\sigma) c(\alpha, \text{power}, \beta_1), \quad v(\sigma) = \frac{(1 - e^{-8\pi^2\sigma^2})e^{4\pi^2\sigma^2}}{8\pi^2}.$$

This assumes equal per-edge noise and edge-disjoint loops. Measured power at the predicted N_{80} is 0.77–0.86.

- The Cramér–Rao bound equals $e^{4\pi^2\sigma^2}/(8\pi^2)$ asymptotically, and the circular mean is 89–100% efficient, so the exponent is intrinsic.

Proposition 15.4 (Discrete groups). *proved; measured looseness* For $\mathbb{Z}/n$ with a symmetric channel, the mode estimator is exact with probability at least $1 - |E|(|A| - 1)(1 - (\sqrt{p_t} - \sqrt{p_w})^2)^m$. The bound holds but is about 10× loose (0.018 against a measured 0.001 at $m = 80$).

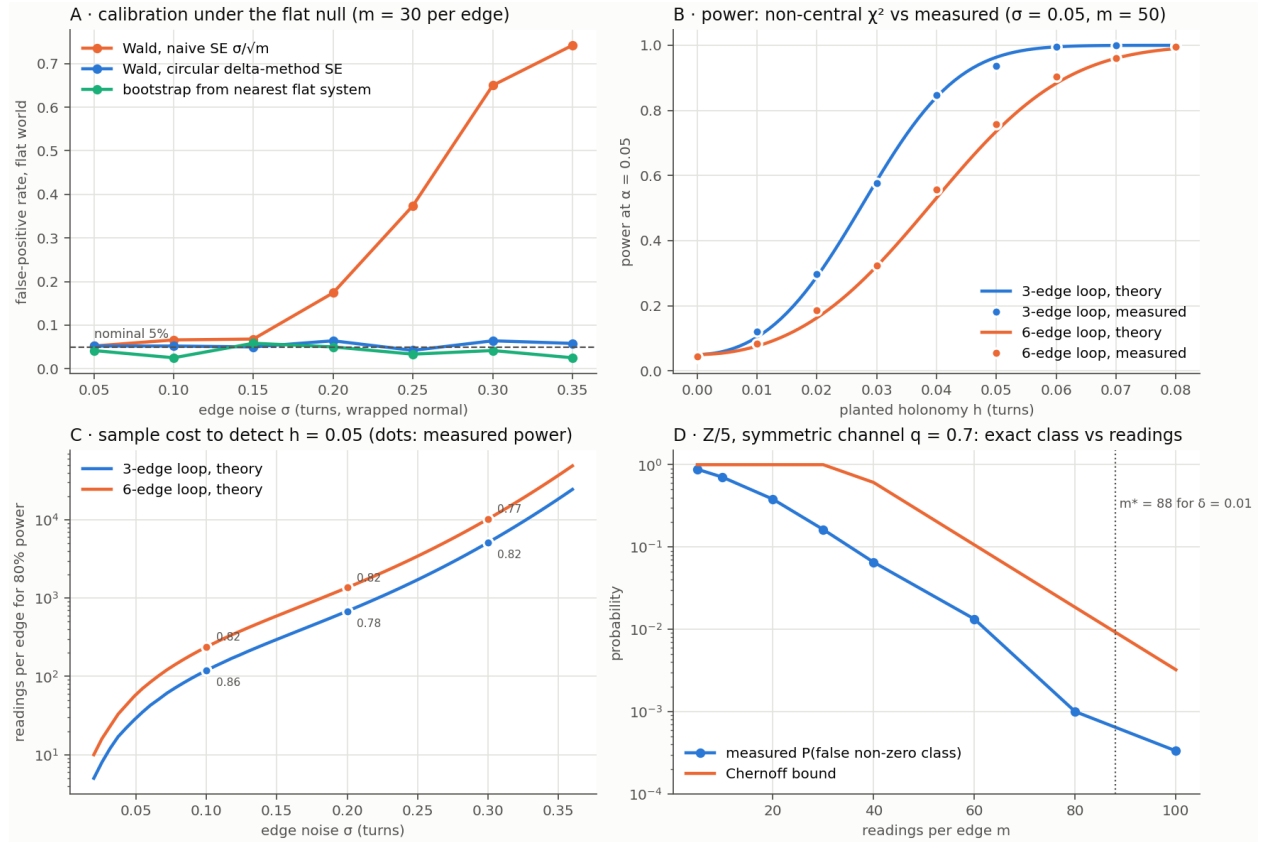


Figure 1: Problem 26 [S]. (A) calibration; (B) power against theory; (C) sample cost; (D) discrete class error against the Chernoff bound.

16 From operations to outcomes, around the probability layer

Theorem 16.1 (The operation-system model e_g). *proved, computed* For $(\Gamma, \mathbb{Z}/n, g)$, let the measurements be the vertex colours, the contexts the edges, and let e_{ab} be uniform on $\{(x, x + g_{ab})\}$. Then:

1. e_g is strongly contextual iff $[g] \neq 0$;
2. for $[g] \neq 0$, $\gamma(s) \neq 0$ for every section over every nonzero coefficient ring;
3. e_g is an instance of an all-versus-nothing model over $\mathbb{Z}/n$ [3].

Proof. (1) A global section is exactly a global field.

(2) Take a compatible R -family with $r_{C_0} = s$. Its restrictions to vertices are functions $f_v : \mathbb{Z}/n \rightarrow R$. Each support is the graph of a bijection, so compatibility forces $f_b(y) = f_a(y - g_{ab})$. Propagating the delta function $f = \delta_{x_0}$ from C_0 around a loop with $h \neq 0$ gives $\delta_{x_0} = \delta_{x_0+h}$, which is impossible when $R \neq 0$. $\square$

Proposition 16.2 (Layer 2 is blind to $[g]$). *proved* In e_g , every edge has $I(c_a; c_b) = \log n$ whatever g is. More generally, any per-context functional invariant under relabelling outcomes (every Shannon or Tsallis functional) is unchanged by post-composing an edge with a permutation.

The ring-invariance in Theorem 16.1(2) contrasts with the mod-3 grid, where γ vanishes over $\mathbb{Q}$. Ring dependence of γ is a property of the family, not of γ .

17 Lossy operations

Each edge carries a doubly stochastic channel K . Colours are uniform, so the Bayesian inverse is $K^\top$. The loop operator M_L is the product around the loop, and the edge-context model e_K has law $\frac{1}{n}K_{ab}$ on edge (a, b) .

Proposition 17.1 (Character eigenvalues). *proved* For covariant channels $C_{\mu_e} S_{g_e}$ on a finite abelian A , every character χ is an eigenvector of M_L with $\lambda_\chi = (\prod_e \hat{\mu}_e(\chi)) \chi(h)$. For the noisy shift $K = (1 - \varepsilon)S_g + \varepsilon U$, $M_L = (1 - \varepsilon)^{|L|} S_h + [1 - (1 - \varepsilon)^{|L|}] U$ and $\lambda_\chi = (1 - \varepsilon)^{|L|} \chi(h)$.

Corollary 17.2. The labelled eigenvalue is base-point free, and its phase is $\chi(h)$ when every $\hat{\mu}_e(\chi)$ is real and positive. The unlabelled multiset is base-point free for any channels, since AB and BA share their nonzero spectrum, but on $\mathbb{Z}/p$ it sees only the order of h . Asymmetric noise adds its own phase: the readout is 2.285 instead of 2.

Proposition 17.3 (Information closed forms). *proved* $I(c_a; c_b) = \log_2 n + (1 - \varepsilon + \frac{\varepsilon}{n}) \log_2(1 - \varepsilon + \frac{\varepsilon}{n}) + (n - 1) \frac{\varepsilon}{n} \log_2 \frac{\varepsilon}{n}$. The loop-composite information is the same with $\varepsilon_L = 1 - (1 - \varepsilon)^{|L|}$. Both are independent of h .

Theorem 17.4 (Contextual fraction on a cycle). *proved* On a cycle of length L with noisy shifts and holonomy $h \neq 0$ in a finite abelian A ,

$$\text{CF}(e_K) = \max\left(0, 1 - \frac{L\varepsilon}{2}\right),$$

independent of $|A|$ and of h .

Proof. Upper bound on NCF. Weight each cell by its discrepancy $d = y - x - g$: $w(0) = 0$, $w(-h) = 1$, and $w = \frac{1}{2}$ otherwise. Discrepancies sum to $-h \neq 0$, so every colouring has weight at least 1. The dual value is $L \frac{\varepsilon}{n} (1 + \frac{n-2}{2}) = \frac{L\varepsilon}{2}$.

Lower bound (packing). Take colourings with one violation of discrepancy $-h$ (mass ε/n per edge), and colourings with two violations $(a, -h - a)$ (mass $\varepsilon/(2n(L-1))$ per ordered edge pair), spread over translations. These fill every off-graph class exactly, and the on-graph load $\frac{L\varepsilon}{2} - \varepsilon(1 - \frac{1}{n})$ fits within the capacity $1 - \varepsilon(1 - \frac{1}{n})$ iff $L\varepsilon \leq 2$.

Beyond the threshold. The model is affine in ε and non-contextual at $\varepsilon = 2/L$ and at $\varepsilon = 1$. $\square$

Theorem 17.5 (Any graph, lower bound). *proved* $\text{CF}(e_K) \geq \max(0, 1 - \ell_{\min}\varepsilon/2)$, where $\ell_{\min}$ is the shortest simple cycle with nonzero holonomy.

Observation 17.6. *measured* Equality holds in 44 configurations: θ -graphs, K_4 , shared edges, shared vertices, disjoint non-flat loops, and $\varepsilon \leq 0.8$. Only $\ell_{\min}$ matters, not the cycle basis. The general packing is open.

Counterexample 17.7 (What does not hold). *computed*

- *Information does not determine CF.* A 3-cycle and a 5-cycle over $\mathbb{Z}/3$ at $\varepsilon = 0.2$ have the same per-edge information, but $\text{CF} = 0.70$ and 0.50 .
- *Non-covariant noise is contextual on its own.* Flat systems reach CF up to 0.24 , and the phase readout of $h = 2$ spreads over $[1.1, 3.1]$.
- *Entropy is not a function of Fourier moduli.* A homometric pair has identical $|\hat{p}|$ but $H = 2.7714$ against 2.7689 bits.

layer	invariant	response to ε	detects h until
1	labelled eigenvalue of M_L	modulus $(1 - \varepsilon)^{ L }$, phase fixed	$\varepsilon \rightarrow 1$
2	per-edge and loop I	decreases, independent of h	never
3	CF	$1 - \ell_{\min}\varepsilon/2$	$\varepsilon = 2/\ell_{\min}$
3	γ	support becomes full	$\varepsilon = 0$ only

Table 9: The same noise acting on four invariants [S].

Proposition 17.8 (Orientation for $\mathbb{Z}/3$ and $\mathbb{Z}/5$). *proved, computed* For odd primes, $h \neq -h$ whenever $h \neq 0$, so the labelled phase always detects orientation. Reversing the loop inverts the readout ($1 \leftrightarrow 4$ and $2 \leftrightarrow 3$ on $\mathbb{Z}/5$). The unlabelled multiset and CF do not detect orientation.

- On $\mathbb{Z}/3$ a non-flat holonomy is exactly a chirality, read from the sign of $\text{Im } \lambda_1$.
- On $\mathbb{Z}/5$ the phase splits into a magnitude class, $\text{Re } \lambda_1/|\lambda_1| = 0.309$ for ± 1 and -0.809 for ± 2 , and an orientation.
- The magnitude class is meaningful only with fixed state labels, since $x \mapsto 2x$ carries $h = 1$ to $h = 2$.

18 The support dial: Tsallis entropies

Proposition 18.1. *computed; the cocycle property is Vigneaux's* Let $S_\alpha = (1 - \sum p^\alpha)/(\alpha - 1)$ with the deformed action $(X \cdot_\alpha F)(P) = \sum_x P(X=x)^\alpha F(P \mid X=x)$.

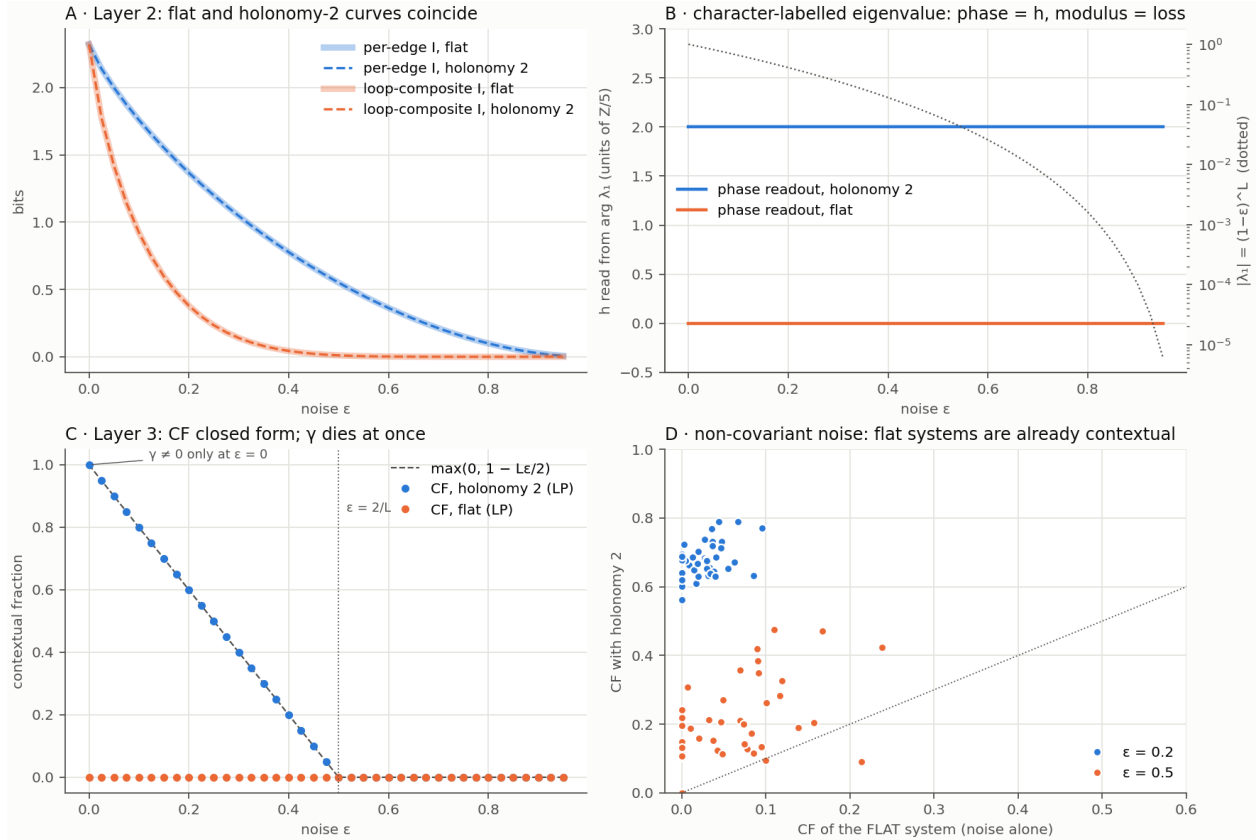


Figure 2: Noisy operations on a 4-cycle over $\mathbb{Z}/5$, with $h = 2$ against a flat system [S].

- S_α is a δ_α -cocycle (defect below 3×10^{-15}), but not a cocycle for Shannon's action (defect up to 1.8).
- $S_0 = |\text{supp}| - 1$, and $S_\alpha \rightarrow H$ as $\alpha \rightarrow 1$.
- Mass ϵ added to an empty cell changes S_0 by 1, and S_α by $\sim \epsilon^\alpha$.
- $S(X) + S(Y) - S(XY)$ reaches -1.71 at $\alpha = 0.5$.

α therefore interpolates between AMB's possibilistic lens ($\alpha = 0$) and Shannon ($\alpha = 1$). Every S_α is symmetric, so Proposition 16.2 still holds.

19 Support strata: where each lens can see anything

An empirical model $e = (e_C)_C$ lies in a product of simplices $\prod_C \Delta(O^C)$, intersected with the no-signalling polytope. Its *support* is $\sigma(e) = (\text{supp } e_C)_C$. The *stratum* of e is the set of models with the same support, that is, the relative interior of a face.

- The *interior* is full support in every context.
- The *vertices* are one outcome per context.

Proposition 19.1 (Support stratification). $[S]_{\text{proved}}$

1. Face functions. *The support presheaf $\mathcal{S}_e$, its global sections, logical and strong contextuality, and γ over every ring depend on e only through $\sigma(e)$. Strong contextuality is inherited by smaller supports: if $\sigma(e) \subseteq \sigma(e')$ componentwise and e' is strongly contextual, so is e .*
2. Interior. *With full support, every local section extends to a global section of $\mathcal{S}_e$. So e is not logically contextual and $\gamma(s) = 0$ for every s , over every ring. The contextual fraction can still be positive.*
3. Vertices. *If each $e_C = \delta_{s_C}$ and e is no-signalling on a cover of the measurements, the s_C glue to a global assignment g with $e = \delta_g|_C$. Then:*
 - *e is non-contextual, $\text{CF} = 0$ and $\gamma = 0$;*
 - *every Shannon and Tsallis functional of every context vanishes, together with every mutual and conditional information, since each is a combination of entropies of point masses.*
4. *The probability lens varies within strata. CF is not a face function. It is convex (Abramsky–Barbosa–Mansfield), hence continuous on the relative interior of the model polytope. S_α for $\alpha > 0$ and H are continuous on closed simplices. $S_0 = |\text{supp}| - 1$ is constant on strata and jumps exactly across them.*

Proof. 1. $\mathcal{S}_e(U)$ is the set of sections over U whose restrictions to each context lie in $\text{supp } e_C$. Every listed invariant is defined from $\mathcal{S}_e$. If $\sigma(e) \subseteq \sigma(e')$ then $\mathcal{S}_e \subseteq \mathcal{S}_{e'}$, so a global section of $\mathcal{S}_e$ is one of $\mathcal{S}_{e'}$.

2. Extend $s \in O^C$ arbitrarily to $g \in O^X$. Every restriction of g lies in a full support, so $g \in \mathcal{S}_e(X)$. A section that extends to a global section has trivial obstruction and is no logical witness.

3. No-signalling gives $\delta_{s_C}|_{C \cap C'} = \delta_{s_{C'}}|_{C \cap C'}$, so $s_C|_{C \cap C'} = s_{C'}|_{C \cap C'}$. Because the contexts cover X , the s_C glue to a unique g , and δ_g is a global law. $H(\delta) = S_\alpha(\delta) = 0$ for all α .

4. The Tsirelson box and white noise lie in the same (interior) stratum, with $\text{CF} = 0.414$ and 0. Continuity of S_α for $\alpha > 0$ and the jump of S_0 are immediate from the formulas. $\square$

Corollary 19.2. *proved A nonzero γ and logical contextuality occur only on intermediate strata: neither full support nor single outcomes.*

- *The PR box has no global section and 8 cohomological witnesses.*
- *The Hardy model has one logical witness, missed by γ over $\mathbb{Z}$.*

Contextuality in the interior is visible only to the probability lens (CF); the possibilistic lens is blind to it. S_0 reads the stratum and $S_{\alpha>0}$ reads the position within it, which completes the support dial of Section 18.

Proposition 19.3 (Two-outcome supports are all-or-nothing). *[S]proved Consider a graph scenario with binary measurements and pairwise contexts. Let e be no-signalling with $|\text{supp } e_C| = 2$ for every context.*

1. *Every measurement is either pinned (deterministic) or free (both values occur). An edge between two free measurements has a bijection support $\{y = x + c_C\}$ with $c_C \in \mathbb{Z}/2$. An edge between a pinned and a free measurement has a product support. No edge joins two pinned measurements.*

2. e is contextual if and only if the $\mathbb{Z}/2$ connection c on the free subgraph has a non-flat cycle. It is then strongly contextual, with $\text{CF} = 1$ and $\gamma \neq 0$. Otherwise every local section extends and $\gamma = 0$. Logical-but-not-strong contextuality, as in Hardy, needs a support of size at least 3.
3. On a component with a non-flat cycle, no-signalling forces every free marginal to be $\frac{1}{2}$, so the stratum is a single point there. On a flat component the stratum is a one-parameter family $q \in (0, 1)$ with $\text{CF} = 0$.
4. Per context, $H = I = h(q)$ and $S_0 = 1$. The non-flat point and the flat point $q = \frac{1}{2}$ have identical per-context Shannon and Tsallis profiles, yet $\text{CF} = 1$ and 0.

Proof. 1. A size-2 subset of $\{0, 1\}^2$ either fixes one coordinate or is the graph of a bijection. No-signalling makes pinned/free a property of the measurement.

2. Global sections are the pinned values together with a flat section of the $\mathbb{Z}/2$ torsor on the free subgraph, which exists if and only if every cycle is flat. If no flat section exists, then $\mathcal{S}_e(X) = \emptyset$, and $\text{CF} = 1$ by Abramsky–Barbosa–Mansfield. If one exists, a torsor section is fixed by any one value on each component, so every local section extends.
3. A bijection edge transports the marginal $p \mapsto p$ or $p \mapsto 1 - p$, and an odd cycle gives $p = 1 - p$.
4. The joint on a bijection edge is determined by its marginal. Relabelling invariance gives the last claim, which is Proposition 16.2 at $n = 2$.

□

So at two outcomes, Layer 3 is exactly $\mathbb{Z}/2$ holonomy. The information lens sees only the free parameter q , and q is pinned to $\frac{1}{2}$ precisely when the holonomy is non-trivial.

Checked: the CHSH parity family (`test_two_outcome_supports_all_or_nothing`):

- flat models are non-contextual for $q = 0.5, 0.7, 0.9$;
- the odd model at $q = 0.9$ signals (defect 0.8);
- at $q = \frac{1}{2}$ the odd model is the PR box, with $\text{CF} = 1$ and a profile identical to the flat model's.

Remark 19.4 (What this replaces). ~~withdrawn~~ A proposed “sheaf-geometric bridge” identified γ_{AMB} with the boundary holonomy of a smooth circle bundle through a support-reduction functor χ . It placed contextuality at single-outcome supports and tied it to the divergence of $\partial H / \partial p_i$ at $\partial \Delta$. It is rejected here, for four reasons:

- Its central equation equates an integer class taken mod $\mathbb{Z}$, which is 0, with a holonomy that is never defined.
- The AMB cover is a finite cover of a measurement set, not an open cover of a manifold, and χ is never constructed.
- Information cohomology uses measurable cochains and functional equations, so the divergence of a derivative cannot create a class. Its $H^{\geq 2}$ is open, and “ $I = dH$ ” conflates de Rham d with δ and δ_t (we have $I = \delta_t H$).
- By Proposition 19.1(3), single-outcome models are exactly where both lenses vanish.

The correct discrete statement relating holonomy and γ is the connecting-class result for operation systems (Theorem 16.1). The correct role of the simplex boundary is the stratification above.

Checked in `tests/test_strata.py`: interior (Tsirelson, noisy PR, noisy GHZ); vertices (random deterministic assignments on the CHSH, GHZ and $\mathbb{Z}/3$ grid scenarios); a CF-versus-face counterexample; and the intermediate faces.

20 Magnitude and phase: what Layer 2 reads

Proposition 16.2 cannot be evaded by choosing a different symmetric functional. This section says what Layer 2 *does* read, and how little must be added to recover CF.

Proposition 20.1 (Logical Bell bound). *[S]proved Let $\varphi_1, \dots, \varphi_n$ be formulas on contexts $C_1, \dots, C_n$ that are jointly unsatisfiable, as in Abramsky [25], and let $p_i = e_{C_i}(\varphi_i)$. Then $\text{CF}(e) \geq \sum_i p_i - (n - 1)$.*

Proof. Write $e = (1 - \lambda)e^{NC} + \lambda e'$ with $\lambda = \text{CF}(e)$. A global assignment satisfies at most $n - 1$ of the φ_i , so the non-contextual part contributes at most $n - 1$ to $\sum_i p_i$, and the rest contributes at most n . Hence $\sum_i p_i \leq (1 - \lambda)(n - 1) + \lambda n$. $\square$

The bound costs linear time. A Liar cycle ($x_1 = x_2, \dots, x_n = \neg x_1$) is the case of odd $\mathbb{Z}/2$ holonomy. For $n = 4$ it is the PR box [25], which is Proposition 19.3.

Proposition 20.2 (CF from moduli and one bit). *[S]lower bound proved; equality computed for $n = 4$. Take a binary n -cycle with uniform marginals and edge correlations $s_i = \mathbb{E}[(-1)^{a+b}]$.*

1. *Layer 2 determines each modulus $|s_i|$ and nothing else about the s_i . The Shannon form is $I_i = \log 2 - h(\frac{1+|s_i|}{2})$. The Tsallis-2 form, with Vigneaux's action, is $I_{2,i} = S_2(b) - a \cdot S_2(b) = (1 + s_i^2)/4$. The signs are invisible by Proposition 16.2, and their product is the holonomy bit.*
2. *The contextual fraction is*

$$\text{CF} = \begin{cases} \max(0, \frac{1}{2}(\sum_i |s_i| - (n - 2))) & \text{odd holonomy,} \\ \max(0, \frac{1}{2}(\sum_i |s_i| - 2 \min_i |s_i| - (n - 2))) & \text{flat.} \end{cases}$$

The right-hand side is a lower bound by Proposition 20.1, applied to the odd sign pattern that is best aligned with the s_i . Equality is computed for $n = 4$: the LP agrees to 10^{-15} on 60 random models in `tests/test_magnitude_phase.py`, and on 300 in an exploratory run.

Consequences.

- The holonomy bit moves CF by at most $\min_i |s_i|$. On 300 random CHSH models it moved CF by 0.078 on average and by 0.82 at most. With a noisy edge, Layer 2 alone nearly determines CF.
- A flat cycle with one uniform edge ($|s_j| = 0$) and the other edges perfect has $\text{CF} = \frac{1}{2}$.
- The Tsallis-2 form needs no inverse entropy: $|s_i| = \sqrt{4I_{2,i} - 1}$. Independent bits give $I_2 = \frac{1}{4}$, not 0, which is non-additivity. S_2 also has an exactly unbiased estimator, $\sum_x n_x(n_x - 1)/N(N - 1)$, so it avoids the Miller–Madow bias of plug-in Shannon estimates.

Remark 20.3 (Grassmannian charts: magnitude is chart-invariant, phase is the chart coordinate). *proved (identities); interpretation*

- *Exponential charts.* $\text{Gr}(k, n) = GL_n/P$ has big cell $\exp(\mathfrak{n}^-)P$ with $\mathfrak{n}^- \cong \text{Hom}(\mathbb{F}^k, \mathbb{F}^{n-k})$. For Grassmannians $\mathfrak{n}^-$ is abelian with $X^2 = 0$, so $\exp X = I + X$ and the chart $X \mapsto \text{row}[I \mid X]$ is linear.

- *Point counts.* Over $\mathbb{F}_q$, $|\mathrm{Gr}(k, n)| = \sum_{\lambda} q^{|\lambda|} = q^{k(n-k)}(1 + O(q^{-1}))$.
- *S_2 is a dimension.* For flags of type $k = (k_1, \dots, k_s)$,

$$\dim \mathrm{Flag}(k) = \sum_{i < j} k_i k_j = \frac{1}{2} \left(n^2 - \sum_i k_i^2 \right), \quad \text{so} \quad S_2(k/n) = \frac{2}{n^2} \dim \mathrm{Flag}(k)$$

exactly. This underlies Vigneaux's limit $\frac{2}{n^2} \log_q \binom{n}{pn}_q \rightarrow S_2(p)$ [6]. Shannon entropy is the log of a set count, the multinomial, and $q \rightarrow 1$ recovers it.

- *Edges as chart points.* For the operation systems over $\mathbb{F}_p^k$, the support of edge e is the graph of G_e , the point $\mathrm{row}[I \mid G_e] \in \mathrm{Gr}(k, 2k)$. Its exponential coordinate is the operation G_e . Affine maps are homogenised to $\mathbb{F}_p^{k+1}$.
- *Two lenses.*
 - Holonomy is the product of coordinates around a loop. The space of global sections has dimension reduced by $\mathrm{rank}(G_{\mathrm{loop}} - I)$; for a translation by $h \neq 0$ the loop matrix $\begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix}$ fixes 1 of 2 dimensions.
 - Dimension, S_2 and entropy are GL -invariant, hence the same in every chart. That is Layer 2's blindness in geometric form.
 - Chart invariants read magnitude (Layer 2). Coordinates composed around loops read phase (Layer 3).
- *Global structure.* No single chart covers $\mathrm{Gr}(k, n)$, and the transitions are non-linear (Plücker). The global invariants, Schubert classes and tautological Chern classes, live in the gluing, just as holonomy does.

Not claimed. Vigneaux proves that differential entropy and dimension are the only degree-1 classes for Gaussian modules without coordinates. Whether dimension is a cocycle in a finite-field information structure built from our operation systems is settled in Proposition 20.4 below.

Proposition 20.4 (Dimension over $\mathbb{F}_p$). *[S]proved; computed Laws on $\mathbb{F}_p^n$ whose support is an affine subspace form a sub-functor: it is closed under linear pushforward and under conditioning on a linear observable. On it, $D[X](P) = \log_p |\mathrm{supp} X_* P|$ is a 1-cocycle for the Shannon action,*

$$D[XY] = D[X] + \sum_x P(x) D[Y](P \mid X = x).$$

The reason is that every fibre $A \cap X^{-1}(x)$ is a coset of $A \cap \ker X$, so all fibres have the same dimension. On general laws the identity fails (defect up to 0.42). The count $S_0 = |\mathrm{supp}| - 1$ is a cocycle for the $\alpha = 0$ action on all laws.

Computed (`magphase.py`, `tests/test_magphase_module.py`): 300 random affine-support laws on $\mathbb{F}_3^3$ with random linear observables give defect $\leq 7 \times 10^{-16}$; general laws give up to 0.42.

Consequence. Dimension is a Layer-2 cocycle in our finite-field setting, on affine supports. Its non-triviality would follow from the Baudot–Bennequin locality convention (local 0-cochains are constants), which we have not checked separately. Per edge it is constant, $D = 1$ for every graph of a bijection of $\mathbb{F}_p$, so it is blind to holonomy, as Proposition 16.2 requires. Holonomy appears only in the dimension of the *glued* section space.

Dimension deficit from estimated supports. [S]_{measured} Setup (`examples/dimension-deficit.py`):

- a 4-cycle over $\mathbb{F}_3$ and $\mathbb{F}_5$, with $N \in \{30, 90, 300\}$ samples per edge and symmetric noise $\varepsilon \in \{0.1, 0.3\}$;
- edge supports thresholded at $\tau \in [0.2, 0.6]$, and an edge accepted only if its support is the graph of an affine bijection;
- the loop map in homogeneous coordinates, and the statistic $\text{rank}_{\mathbb{F}_p}(M_{\text{loop}} - I)$;
- a null band from a parametric bootstrap under the nearest flat system.

Three systems are planted:

- *flat*: deficit 0, p global sections;
- *translation holonomy*: deficit 1, 0 global sections, strongly contextual;
- *multiplicative holonomy* ($a_{\text{loop}} = 2$): deficit 1, exactly 1 global section (the fixed point), logically but not strongly contextual.

Findings:

- Whenever all supports were recovered, the deficit was correct in every replicate. The null band was 0 at every (ε, N, τ) .
- The statistic fails by *abstaining*: at $\mathbb{F}_5$, $\varepsilon = 0.3$, $N = 30$, recovery is 2–9%.
- γ over $\mathbb{Z}$ and over $\mathbb{Z}/p$ agrees with the deficit in all six cases. The holonomy test covers only translations.
- The global-section count separates strong from logical contextuality, which the deficit alone does not.

Scope. Under symmetric noise a failed recovery is a non-bijection, never a wrong bijection, so the null band is trivially 0. Non-covariant noise could produce wrong bijections and is untested.

Estimation: S_2 versus Shannon. [S]_{measured} Moduli $|s_i|$ and CF of a binary 4-cycle were estimated from N samples per edge, with 400 replicates for each of three configurations (`examples/s2-vs-shannon.py`). Four estimators were compared: unbiased I_2 , Shannon plug-in, Miller–Madow, and the labelled sample correlation.

- At $N \geq 100$ all four agree to within 0.003 in RMSE of CF.
- At $N = 30$ the S_2 route has the smallest bias in s^2 on weak edges (0.017 against 0.035). It has the *largest* CF RMSE on strong and near-threshold edges (0.087 against 0.067, and 0.153 against 0.125), because CF needs $|s| = \sqrt{s^2}$ clipped to $[0, 1]$.
- S_2 *does not pay off here*. With uniform marginals, each 2×2 table has a one-dimensional sufficient statistic, the agreement count. Every lens is then a function of it, and unbiasedness of s^2 does not carry through the square root.

21 A contradiction meter with provenance

Input. A claim (u, i, j, r) means that provenance unit u (a source, document or turn) asserts $x_i \oplus x_j = r$ about binary propositions. Claims may be single assertions (deterministic mode) or repeated samples.

Three separate failures [S]:

1. *Self-contradiction*: an odd cycle among one unit’s own claims.
2. *Overlap disagreement* ($d^0 \neq 0$, the deficit): two units give the same edge different parities, significant at $z > 3$ in sampled mode. It is recorded and the edge is *not* glued.
3. *Contextuality across provenance* ($d^0 = 0$, $H^1 \neq 0$): the overlaps agree, but the glued graph has an odd cycle contained in no single unit. This is a Liar cycle (Proposition 19.3).

Each cycle is reported with:

- the logical-Bell lower bound on CF (Proposition 20.1);
- a union-bound confidence that all of its parities are right;
- a blamed edge, only when every other edge comes from a declared trusted unit.

Coherent fabrications are invisible by construction.

v1 changes [S]:

1. *Pooled noise*. The flip rate $\hat{\varepsilon}$ is estimated by EM over all claims, with each edge’s true parity latent. An edge’s error probability is then the posterior $\lambda^d/(1 + \lambda^d)$, where $\lambda = \hat{\varepsilon}/(1 - \hat{\varepsilon})$ and d is the majority margin. This replaces the worst-case binomial.
2. *Shortest odd cycles*. For every edge, the meter finds the shortest odd cycle through it by BFS in the parity double cover. It replaces fundamental cycles.
3. *Multiple testing*. Cycles are certified at level α/k over the k candidates (Bonferroni).

Benchmark measured (`examples/meter_benchmark.py`).

- *Data*: 12 propositions and 4 honest units asserting 6 true parities each, with one added unit per scenario; 200 instances per cell. Results are in Table 10.
- *Pooling*: it fixes the $m = 5$ row. At $\varepsilon = 0.1$, self-contradictions go from 0.00 to 0.78 and cross-unit cycles from 0.33 to 0.84.
- *Clean cases*: honest-only firing stays at or below 0.04, and coherent fabrications are not flagged.
- *Limits*: at $\varepsilon = 0.2$ with $m \leq 5$, power is low.
- *Validation*: `tests/test_meter.py` checks the CF lower bound against the LP on 3- and 5-cycles.

m, ε , noise model	honest	self	overlap	cross (correct edge blamed)	fabrication
1, 0 (deterministic)	0.00	1.00	1.00	1.00 (1.00)	0.00
5, 0.1, worst case	0.00	0.00	0.74	0.33 (0.32)	0.01
5, 0.1, pooled	0.01	0.78	0.86	0.84 (0.84)	0.01
3, 0.1, pooled	0.04	0.37	0.61	0.55 (0.55)	0.04
15, 0.2, pooled	0.01	0.95	0.98	0.94 (0.94)	0.00
5, 0.2, pooled	0.04	0.21	0.56	0.19 (0.19)	0.04

Table 10: Detection rates by planted scenario. The “honest” column is the false-positive rate of the full meter.

General binary claims. [S]proved (classification); computed Each unit’s formulas define a local support $S_u \subseteq \{0, 1\}^{C_u}$ (`meter_general.py`; formula syntax: $\sim, \&, |, \wedge, \rightarrow, \leftrightarrow$). A deletion-based minimal unsatisfiable set of units is classified as one of:

- *self*: $S_u = \emptyset$;
- *overlap*: two units with disjoint projections onto their shared propositions;
- *propagation*: arc consistency (semijoin reduction) empties a unit;
- *cyclic*: arc consistency leaves every S_u non-empty and consistent on overlaps ($d^0 = 0$), yet no global assignment exists.

By Beeri–Fagin–Maier–Yannakakis [8], the cyclic type requires a cyclic hypergraph; this is the strongly contextual case. With assertion frequencies p_u , the core carries the logical-Bell bound $CF \geq \sum p_u - (k - 1)$.

Gaussian fast path over $\mathbb{F}_2$. [S]computed Weights of Gaussian type are closed under gluing, and gluing them is matrix algebra rather than a sum over assignments. The Grassmann–Gaussian weights of arXiv:1305.3246 are an example of this principle; we read only its abstract. Over $\mathbb{F}_2$ the analogue is a unit whose support is an affine subspace, such as a parity, XOR or literal claim or a conjunction of these.

- *Method*: for such units, contradiction cores come from Gaussian elimination with row tracking. The combination y with $yA = 0$ and $yb = 1$ is the certificate. A core is already minimal when $\text{rank} = |\text{rows}| - 1$ (a circuit); otherwise it is minimised by deletion.
- *Timing*: brute force takes 0.35 s at 16 propositions and grows as 2^n . The fast path handles a 5,000-claim Liar ring among 15,000 consistent distractor claims (20,000 propositions) in 0.66 s.
- *Validation*: cores match brute force and are minimal on 40 random parity systems (`tests/test_meter_general.py`).
- *Scope*: non-affine claims such as $A \rightarrow B$ keep the exact but exponential path. General satisfiability is NP-hard, and the CF LP is not reduced by this.

First real run: BASIL, overlap channel. [S]measured **Setup** (`examples/basil_overlap.py`):

- BASIL [26]: 97 events with all three outlets (Fox, NYT, HuffPost) annotated; gold annotations only.
- Sources are the outlets. A proposition is “the article is negative toward target T in event t ”.
- Literals are encoded as parity edges to an anchor proposition, so cycles cannot arise and only the overlap channel is exercised.
- Null: outlet labels permuted within events, 1000 permutations.

reading	Fox–NYT	Fox–HPO	NYT–HPO	separation (p)
A: article-level author feeling (deterministic)	12/26	6/17	1/27	0.370 (0.002)
B: phrase-level spans (sampled, $\hat{\varepsilon} = 0.114$)	11/112	9/115	7/113	0.026 (0.21)

Table 11: Overlap disagreements over shared propositions. Separation is the mean Fox-versus-others rate minus the NYT–HPO rate.

- At the article level Fox separates from the other two ($p = 0.002$). In span-level polarity it does not ($p = 0.21$).
- Disagreeing article pairs have different annotated relative stance in 19/19 (A) and 26/27 (B) cases, against a base rate of 240/291. The binomial tails are 0.026 and 0.037.
- *Caveats.*
 - Stance and feelings come from the same annotators, so the stance check is not independent.
 - The counts are small: 17–27 shared propositions in reading A.
 - These disagreements are differences in slant, recorded as overlap. They are neither factual contradictions nor hallucinations.
 - Cycles need relational claims, which BASIL’s annotations do not provide.

One example in `tests/test_meter_general.py` matters for the attention proposals reviewed earlier: “ $A \rightarrow B, B \rightarrow C, C \rightarrow \neg A$ ” is *satisfiable* (take A false), so it is not a contradiction. Adding the claim “ A ” produces a propagation-type core. The parity version, $A \leftrightarrow B, B \leftrightarrow C, C \leftrightarrow \neg A$, is cyclic.

22 The decorated Bregman–Čech nerve

Tokens are points $p_i \in \Delta(\text{states})$ with a time τ_i . The radius of a simplex is $\min_c \max_i D_{\text{KL}}(p_i \| c)$.

Lemma 22.1. *proved, computed* The balls $\{p : D_{\text{KL}}(p \| c) \leq r\}$ are convex, since KL is convex in its first argument, so the nerve lemma applies. For a pair, the optimum lies on the segment between the two points; it is found by bisection and certified by the minimax dual $\max_\lambda [\lambda D_{\text{KL}}(p \| m) + (1 - \lambda) D_{\text{KL}}(q \| m)]$ to 10^{-9} .

Construction 22.2 (Decoration). • Edges are oriented by time (“A before B”) and labelled by the best-aligning group element, with its margin over the runner-up.

- Triangles carry their boundary holonomy; a nonzero value is a non-flat event.
- Each H_1 bar is labelled by the holonomy of its birth cycle.

Proposition 22.3 (Invariants for $\text{Aff}(\mathbb{Z}/p)$). *proved* Loop labels depend on the base point only by conjugation. The conjugacy classes are: the identity; all nontrivial translations, as one class; and one class per multiplier $u \neq 1$. For odd p , $\text{Aff}(\mathbb{Z}/p)^{ab} = (\mathbb{Z}/p)^\times$, so only u is a function of the H_1 class. Relabelling tokens preserves the class and not the element (tested).

Proposition 22.4 (Walls). *proved; consequence measured* A permutation σ of n states fixes a face of codimension $n - \#\text{cycles}(\sigma)$. In $\text{Aff}(\mathbb{Z}/3) \cong S_3$ the reflections are transpositions, fixing codimension-1 walls. The quotient is a Coxeter chamber, and every planted holonomy is absorbed into a flat labelling (0/100 recovered). Every non-identity element of $\text{Aff}(\mathbb{Z}/5)$ has codimension ≥ 2 .

Observation 22.5. *measured* On $\text{Aff}(\mathbb{Z}/5)$, rings with every edge margin $\geq 10^{-3}$ are recovered in 254/255 cases; rings with an ambiguous edge are wrong in 57/145. On $\mathbb{Z}/3$ translations, both orientations are recovered and reversing time inverts the label.

Limits *measured*:

- Dirichlet noise on tokens (concentration $2000 \rightarrow 200$) raises ring-label error from 15% to 48–63%.
- The $\mathbb{Z}/2$ barcode readout is confounded by shortcut loops of the quotient and by coefficients: 18/20 correct on $\mathbb{Z}/5$ translations and 8/12 on $\mathbb{Z}/3$.
- In the non-abelian case, bars with non-identity labels can die at flat triangles.

Semantics adopted. Negation is *partial*: it is undefined on U and sends U to an explicit *not-applicable* state. Operations therefore form an inverse monoid, not a group, and are lossy.

23 Two facets: restriction and extension

Let $i : \Sigma \rightarrow \mathcal{S}$ include the cover into the information site.

Proposition 23.1. *proved, computed*

1. Restriction $R: P \mapsto (\pi_C P)_C$ has image exactly the non-contextual models.
2. Extension $E: e \mapsto \text{Ext}(e) = \{P : R(P) = e\}$ is a convex polytope, nonempty iff $\text{CF}(e) = 0$. If $\text{CF} = 0$, there is $b \geq 0$ with $\sum b = 1$ and $R(b) \leq e$, and equal context totals force $R(b) = e$.
3. $P \in E(R(P))$, and $R(E(e)) = \{e\}$ whenever $E(e) \neq \emptyset$.
4. The canonical member of $E(e)$ is the maximum-entropy extension, found by iterative proportional fitting (Csiszár). IPF fails to converge on contextual models.

Observation 23.2 (Graded extension). *measured* The optimal non-contextual parts, of mass $1 - \text{CF}$, form a face. In CHSH it is a single point for every contextual model tested (39/39); in the three-party GHZ scenario it never is (0/10). The maximum-entropy point of the face gives a unique representative.

These maps act on models, not on cohomology, so they are consistent with Theorem 13.1. The attached drafts' forward map, which sent γ/CF to the spectral modulus, is not a functor.

24 Not-applicable entries: the gated test

Records are windows of labels (T, F, could-T, could-F, U, n/a), with a *mask* marking variables that are not in the context. The mask is distinct from the *n/a value*.

Construction 24.1 (Pipeline). 1. GYO reduction of the maximal observed contexts. If the cover is acyclic, the pipeline stops (Theorem 10.4); any $\text{CF} > 0$ is then inconsistency, not contextuality.

2. Possibilistic test across support thresholds, with its null band.
3. CF by LP, with the signalling defect reported beside it.
4. Null: each record keeps its mask, and its values are redrawn from the maximum-likelihood consistent law. That law is found by missing-data EM (Theorem 10.2) and pinned to maximum entropy.

Proposition 24.2 (The deficit). *proved (zero set); measured (the rest)* The deficit is the per-record log-likelihood lost by the best consistent law, $\min_P \sum_C w_C D_{\text{KL}}(e_C \| P_C)$.

stream	gate	CF	verdict
random N/A, complete records present	acyclic	—	stop: the AMB half is empty
honest, one N/A per record	cyclic	0.029	null mean 0.027, $p = 0.39$: stop
structured, shift twist	cyclic	0.858	above all 30 null replicates (max 0.044)
structured, negation with $U \rightarrow n/a$	cyclic	0.859	above the null, but signalling 0.18

Table 12: The gated test on 9,000-record streams [S].

- Zero set. *For no-signalling tables it vanishes iff $CF = 0$: the minimum is attained, and KL is zero iff the tables coincide.*
- Not a function of CF . *At $CF = 0.5$ on noisy-shift cycles it is 0.101, 0.068, 0.051 for $L = 3, 4, 5$.*
- Closed form at $\varepsilon = 0$: *it equals $\log(L/(L - 1))$ (measured to 10^{-4}).*
- Sampling floor. *On honest data it falls as about $1/N$ (0.0079, 0.0010, 0.00015 for $N = 10^3, 9 \cdot 10^3, 8.1 \cdot 10^4$); on contextual data it stays flat at about 0.282.*
- Bootstrap intervals: *[0.270, 0.287] for the shift twist and [0.291, 0.309] for the negation twist.*

Observation 24.3 (The ridge). *measured* When contexts are pairs, the three-way interaction is unidentified. EM from different starts reaches the same likelihood with laws 0.005–0.023 apart in L_1 . This is the upper-level identifiability failure of Part I in three variables. The maximum-entropy pin removes the start dependence (tested); without it, a null on larger contexts would measure ridge drift.

Observation 24.4 (Thresholded γ). *measured* Near a cell’s expected mass, sampling flips cells in and out of the support and manufactures logical witnesses. The null fires in 20/20 replicates at threshold 0.005 on honest data, and at 0.1 on the contextual streams’ null. γ is admissible only where the null band is near zero: at 0.02 and 0.15, both structured streams give $\gamma \neq 0$ at 5/5 against a null of 0/20. Report the full sweep.

25 O2: when is a merged model minimal sufficient?

Fix a level l of a binary tree grammar and one node n at that level. Write $B \in [V]$ for its symbol, X for the leaves of its subtree and O for all other leaves. The grammar gives $X \perp O \mid B$, so

$$P(x, o) = \sum_b P(b) D[b, x] U[b, o], \quad D[b, x] = P(x \mid b), \quad U[b, o] = P(o \mid b).$$

A *partition merge* $f : [V] \rightarrow [V']$ replaces B by $f(B)$ and pools rows with weights $P(b \mid f(B))$. It is *exact* if $X \perp O \mid f(B)$, so the merged model reproduces the law of (X, O) .

Proposition 25.1 (Lossless merges). *[S]proved* A block S of f is exact if either

- (i) *the downward rows $D[b]$, $b \in S$, are equal (in particular, if the child rows $R_l[b]$ are equal), or*
- (ii) *the upward rows $U[b]$, $b \in S$, are equal.*

Proof. The block contributes $M_S = \sum_{b \in S} P(b) D[b] U[b]^\top$ to $P(X, O)$. Under (i), $M_S = d_S (\sum_{b \in S} P(b) U[b])^\top$. Under (ii), $M_S = (\sum_{b \in S} P(b) D[b]) u_S^\top$. Either way $M_S = P(S) P(X \mid S) P(O \mid S)^\top$. $\square$

Case (ii) is exact at a position. In a grammar with one table per level it needs $P(b \mid f(B))$ to be the same at every node of the level. This holds automatically at the root, which is a single node.

Proposition 25.2 (The minimal sufficient statistic is a rank object). *[S]proved*

1. The minimal sufficient statistic of X for O is $x \mapsto P(O \mid x) = q(x)^\top U$, where $q(x) = P(B \mid x)$. Equivalently it is $q(x)$ modulo $\ker U^\top$.
2. Its values span a space of dimension $r^* = \text{rank } P(X, O) = \text{rank}(G^{1/2} K^{1/2})$, where $G = DD^\top$ is computed by the recursion $G_l = R_l(G_{l-1} \otimes G_{l-1})R_l^\top$ and K is the outside Gram.
3. Any exact merge has $V' \geq r^*$.

Proof. 1. $P(o \mid x) = \sum_b P(b \mid x)U[b, o]$ by the Markov property. Minimality is the standard likelihood-ratio characterisation.

2. $P(X, O) = \text{diag}(p_X)[P(O \mid x)]_x$, so the span of the $P(O \mid x)$ has dimension $\text{rank } P(X, O)$. Write $D = G^{1/2}W$ and $U' = K^{1/2}W'$ with W, W' row-orthonormal on the relevant ranges. Then $P(X, O) = W^\top G^{1/2} K^{1/2} W'$.

3. An exact merge factors $P(X, O)$ through V' states.

□

Theorem 25.3 (When a partition reaches the minimum). *[S]proved* Suppose the rows $D[b]$ are linearly independent.

1. A block S is exact if and only if $U[b]$ is constant on S .
2. A partition merge attains $V' = r^*$ if and only if the number of distinct upward rows equals $\text{rank } U$. That is, the whole rank deficiency of U must come from equal rows.

Proof. With D_S the matrix of rows $D[b]$, $b \in S$, we have $M_S = D_S^\top W_S$ with W_S having rows $P(b)U[b]$. Since D_S has independent rows, $\text{rank } M_S = \text{rank } W_S$. This is 1 if and only if the $P(b)U[b]$ are proportional, that is, the probability vectors $U[b]$ are equal. The coarsest exact partition is therefore “equal upward rows”. Its size is the number of distinct rows, while $r^* = \text{rank } D^\top \text{diag}(P(b))U = \text{rank } U$. □

Counterexample 25.4. *computed* Take $V = 3$ with independent D , and $U[c] = \frac{1}{2}(U[a] + U[b])$ with $U[a] \neq U[b]$. Then $r^* = 2$, but none of the three 2-block partitions is exact. The minimal sufficient statistic $q_a + \frac{1}{2}q_c$ mixes symbols fractionally, so no merge can produce it. This is checked in `tests/test_o2.py`.

Proposition 25.5 (Bregman cost is a complete-data quantity). *[S]proved* The split-merge cost of merging a block S is $\sum_{b \in S} n_b \text{KL}(R_l[b] \parallel \bar{R}_S) = n_S I(B; \text{children} \mid B \in S)$. It vanishes if and only if case (i) holds. It is blind to case (ii): an upward-exact merge has zero observed-data loss but positive cost whenever the child rows in S differ.

Proof. The identity is the Bregman-information form of the mutual information, as in Banerjee et al. [16]. Zero-loss under (ii) is Proposition 25.1. □

Corollary 25.6 (The root collapses). *proved* The root has an empty outside, so U has rank 1 and every root partition is exact. The root can always be replaced by one symbol with coupling $\mu(u, v) = \sum_b \pi_b R_L[b, u, v]$, since $P(x) = \sum_{u, v} \mu(u, v) \beta_u \beta_v$. The complete-data cost of this collapse is $I(B_{\text{root}}; \text{children}) > 0$.

level	reachable V	distinct child rows	rank D	distinct upward rows	r^*
1 (left, right)	7	7	7	7	7
2 (left, right)	8	8	8	8	8
3 (left, right)	8	8	8	8	8
4 (root)	8	8	8	1	1

Table 13: O2 ranks on the true grammar of Corpus B (`examples/o2_ranks.py`).

Corpus B, true grammar. `computed` Corpus B was regenerated with the builder’s flags. The rules are identical, because they are drawn before sampling.

Level-1 symbol 2 is never produced by a level-2 rule. A first pass that kept it showed $r^* = 7 < 8$ with 8 distinct rows. That is unreachability, not a non-partition deficiency.

By Theorem 25.3:

- At levels 1–3 no nontrivial exact merge exists, and the reachable true grammar is already minimal sufficient.
- At the root the minimum is 1.

On the regenerated validation split (2000 sequences):

- the 8-symbol root and the single μ -root give identical held-out $\log p$ per sequence, -15.282238 (difference 0 to machine precision);
- the complete-data cost of the collapse is 1.799 nats per root node;
- across all 127 two-block root partitions the cost ranges over $[1.106, 1.563]$ nats, while the observed loss is exactly 0 for every one.

This explains the level-4 anomaly in [H]: merging $8 \rightarrow 2$ at the root cost 0.72 nats/node yet improved held-out likelihood. Proposition 25.5 predicts exactly this, and it is why split–merge must accept merges on held-out likelihood rather than on cost.

Status of O2.

- Settled for the true grammar: the characterisation (Theorem 25.3), the bound $V' \geq r^*$, and the Corpus B verdict.
- Open: the same test on fitted tables, where exact equalities become statistical ones. A test for equal upward rows needs an outside-Gram estimator and a null, in the same pattern as V6.
- Open: whether real corpora exhibit Counterexample 25.4. That would require a soft, non-partition compression.

26 Experiment: grammar-guided transformers on Corpus B

Question. Does the split–merge grammar help a transformer beyond what the transformer learns alone?

Setup [S] `measured`.

- Corpus B: 19,000 training sequences, 1,000 for early stopping, and the 2,000-sequence validation split.

- The grammar is fitted by split–merge (3 rounds) and scores -15.421 against the truth’s -15.282 .
- The model is a causal transformer with $d = 64$ and 4 heads, trained for 4,000 AdamW steps at batch size 256.
- The guidance is the fitted grammar’s prefix posteriors $P(B_\ell(t) \mid x_{<t})$ of the four ancestors of the next leaf, computed by inside–outside with unobserved leaves set to 1.
- Variants:
 - *base*: no guidance;
 - *aux*: auxiliary heads distil the posteriors, and nothing is used at test time;
 - *feed*: the log-posteriors are added to the input embedding, so the grammar is used at test time.
- *Oracle* rows use posteriors from the true grammar.
- The metric is held-out excess NLL over the true grammar, in nats per sequence. It is split by the level of the lowest common ancestor of leaves $t - 1$ and t .

model	total excess	lvl 1	lvl 2	lvl 3	lvl 4
fitted grammar alone	0.139	0.028	0.096	0.011	0.004
base, 2 layers (seeds 0, 1; 8k steps)	0.454, 0.490; 0.464	0.14–0.17	0.18–0.19	0.12	0.01–0.02
base, 4 layers	0.369	0.129	0.125	0.090	0.023
aux, 2 layers (seeds 0, 1)	0.267, 0.299	0.07–0.08	0.11–0.13	0.06–0.08	0.01–0.02
aux, 4 layers	0.274	0.084	0.113	0.057	0.014
feed, 2 layers (seeds 0, 1)	0.145, 0.093	0.02–0.04	0.03–0.06	0.03	0.01–0.02
feed, 4 layers	0.099	0.024	0.034	0.031	0.008
oracle aux / oracle feed, 2 layers	0.269 / 0.132	—	—	—	—

Table 14: Excess NLL over the true grammar on 2,000 validation sequences. The standard error of a total is about 0.01–0.02.

Findings.

- *The baseline is data-limited, not undertrained.* Doubling the steps changes nothing ($0.454 \rightarrow 0.464$). Its largest deficits are at the level-2 and level-3 boundaries, where the fitted grammar’s level-3 excess is 11 times smaller.
- *Distillation helps without using the grammar at test time.* The paired difference is -0.187 ± 0.021 at 2 layers and -0.096 ± 0.021 at 4 layers, and the level-3 deficit falls by about 40%. Oracle targets give the same result (0.269). So the gain is not limited by the quality of the fit.
- *Feeding the posteriors reaches the grammar, not beyond.* Against the grammar alone the paired differences are -0.046 ± 0.013 , $+0.006 \pm 0.014$ and -0.040 ± 0.013 . Even oracle posteriors leave 0.13 nats, which is the cost of the transformer’s read-out.
- The transformer improves on the grammar’s level-2 error and loses at levels 3–4.

Scope.

- One corpus in the identifiable regime (781 observations per parameter at level 3).

- One fit; two seeds at 2 layers and one at 4.
- Corpus A is untested. There the fit itself fails at level 3, and guidance cannot supply what the fit lacks.
- Not tested: the specific proposal of Bregman-merge E-steps on hidden states with a multiplicative attention mask. Section 25 argues against its merge criterion.

27 Experiment: recursive training (model collapse) on Corpus B

Question. Can the support dial serve as a label-free collapse metric? Does the small- α lens see collapse before Shannon entropy or likelihood?

Setup [S] measured.

- Generation 0 is the split-merge fit on real data.
- Each generation samples N sequences from the current model and refits by EM (30 sweeps, warm-started).
- Three arms:
 - *replace*: train on the newest synthetic sample only;
 - *accumulate*: train on real data plus all synthetic samples so far;
 - *fresh*: each generation trains on fresh real data. This is the null, and it has the same retraining noise with no feedback loop.
- Metrics, computed on 20,000 model samples of 8-leaf blocks:
 - S_α for $\alpha \in \{0, 0.5, 1, 2\}$;
 - recall of the true block support and of its rarest half, the tail;
 - held-out log-likelihood of *real* data.

$N = 1000$, generation 20	$\log p$ (real)	S_0 of 8-blocks	Shannon H	tail recall
generation 0	−15.421	4206	8.072	0.977
replace	−17.266	3716 (−11.7%)	7.801 (−3.4%)	0.842
accumulate	−15.453	4141	8.041	0.970
fresh (null, generation 19)	−15.451	4185	8.058	0.974

Table 15: Collapse under replacement, with none under accumulation or fresh data. At $N = 4000$ replacement collapses more slowly: $\log p$ goes from −15.421 to −15.629 and tail recall from 0.977 to 0.972 after 20 generations, while accumulation stays at −15.434. At $N = 20,000$, six generations move $\log p$ by only 0.011.

Which lens first? A generation is flagged when its change from generation 0 falls below −3 times the RMS drift of the fresh arm (`collapse_analysis2.py`).

- Under replacement, S_0 , $S_{0.5}$, H , S_2 and $\log p$ on real data are *all* flagged at generation 3. The accumulate arm is flagged 0 of 20 times by every metric.
- At generation 20 the z values are: S_0 −15.0, $S_{0.5}$ −14.6, H −15.0, S_2 −18.6, $\log p$ −48.5, tail recall −18.5.

- The largest *relative* change is in S_0 (−11.7%, the fraction of distinct block types lost). This matches the tail-recall loss (−13.8%) and the likelihood loss.
- *The hypothesis that the support lens detects collapse earlier is not supported.* Every order of α detects at the same generation. S_0 is the most interpretable reading and S_2 the most precise.
- A naive null that uses sampling noise alone, with no refitting, makes H and S_2 flag the harmless drift of the accumulate arm. The retraining null is necessary.

Scope.

- One synthetic grammar and one generation-0 fit. The generation-0 model carries spurious mass (precision 0.963), and early generations shed some of it.
- This is a corpus-level signal, not per-document detection of machine text.

28 Experiment: unknown branching (Corpus V)

Grammar [S]. Every production is $b \rightarrow c_0 c_1 \cdots c_k$: one left child, and $k \in \{0, \dots, 5\}$ right children. The depth is fixed ($L = 3$), with $V = 8$ symbols, $A = 12$ leaves and $m = 3$ productions per symbol. Each production draws its own k , so sentence length varies (mean 77, range 3–149) and the tree is hidden.

Exact inference. Store each symbol as a span matrix, $B[i, j] = P(\text{symbol yields } x_{i:j})$.

- Concatenating children is a matrix product: $B_l[b] = \sum_r p_r B_{l-1}[c_0] \cdots B_{l-1}[c_k]$.
- A sparse chart version gives exact inside–outside in 1.7 ms per sentence.
- It agrees with brute-force enumeration to 10^{-16} , and expected counts agree with finite differences.
- Every validation sentence has exactly one derivation under the true grammar. The true log-likelihood is −30.87 nats per sentence (0.40 per token).

Learners (head plus right children; banded span tables; EM with autodiff expected counts; checked against brute force):

- *Markov*: $k \sim K[b]$, $c_0 \sim H[b]$, $c_t \sim R[b, c_{t-1}]$.
- *Positional*: $k \sim K[b]$, $c_0 \sim H[b, k]$, $c_t \sim R[b, k, t]$. It contains the true grammar except where a symbol has two productions with the same k ; there are 13 such collisions over the three levels.

model (1000 training sentences)	held-out $\log p$ / sentence
true grammar	−30.87
positional, from gold trees (an EM fixed point)	−42.71
Markov, from gold trees (3 EM sweeps: −82.19)	−82.56
positional, EM from random, 20 sweeps (seeds 0, 1)	−82.80, −90.43
bigram / unigram	−151.9 / −189.2

Table 16: Corpus V *measured*.

Chunks and split–merge *measured* (`vchunks.py`, `vsplitmerge.py`).

- *Chunk induction*, level by level:

1. candidate substrings of length at most 6;
 2. EM for a unigram chunk model over segmentations;
 3. pruning of chunk types that are concatenations of other types, which counters the bias toward under-segmentation;
 4. spectral clustering by neighbouring chunks.
- *Level 1*: all 24 true productions are recovered, with token precision 1 and cluster purity 1.
 - *Level 2*: all 24 productions are recovered with precision 1, but one true symbol is split across two clusters and two true symbols share a cluster.
 - *The grammar*: productions are read off the chunks, and the root is collapsed to one symbol, which is lossless by the root corollary of Section 25. It scores -31.77 , parses every held-out sentence, and EM leaves it unchanged.
 - *Split-merge*: splits and merges of symbol assignments, accepted only when exact held-out likelihood improves. Three moves bring it to -30.91 , against the truth’s -30.87 .
 - *Residual error*: every final cluster is pure, but one true symbol stays split in two. At 500 held-out sentences, likelihood does not separate this nearly lossless split, as Section 25 would predict. A parsimony tie-break would settle it.

learner (1000 training sentences)	held-out $\log p$ / sentence
true grammar	-30.87
chunks + split-merge	-30.91
chunks only	-31.77
positional, from gold trees	-42.71
positional, EM from random	-82.80

Table 17: Corpus V: bottom-up chunk induction removes the basin problem.

Findings.

- *Model class*. A production’s children form a fixed tuple, so a first-order chain loses 51.7 nats per sentence even with gold trees. Conditioning on arity and position loses 11.8, which comes from the arity collisions.
- *Search*. EM from random starts climbs well past the bigram, but after 20 sweeps it sits about 40 nats below the best model in its own class. That is the basin-selection problem of Corpus B again, and it motivates split-merge or chunk-based initialisation.

29 Monodromy: representations of π_1 and twisted cohomology

Construction [S] (`monodromy.py`). Transport along an edge $a \rightarrow b$ is the conditional expectation $T_{ab}[x, y] = P(b = y \mid a = x)$. Composing around the fundamental cycles gives

$$\rho : \pi_1(\Gamma) = F_{\beta_1} \rightarrow \text{End}(\mathbb{C}^n).$$

It is unitary exactly when every edge is a bijection; lossy edges give contractions.

Proposition 29.1 (What the representation adds). *proved; computed*

1. Abelian holonomy (shifts on $\mathbb{Z}/n$): $\rho(\ell_i) = T_{h_i}$. The representation is determined by the class $[g]$, and in the character basis it is $\text{diag}(\omega^{kh_i})$, a pure phase with no permutation part. Reversing a loop inverts the phases.
2. Non-abelian holonomy ($\text{Aff}(\mathbb{Z}/p)$): ρ sees more than the abelianised class. Two systems with the same multiplier class on each loop can have different traces and different commutators; see the test with loop traces $(0, 1)$ against $(5, 1)$. A multiplier -1 acts on characters as the reflection $\chi_k \mapsto \chi_{-k}$. Translations do not.
3. Twisted cohomology. Twist the coboundary by the character k : $(\delta f)_e = f(b) - \omega^{kg_e} f(a)$. On a connected graph, $h_k^0 = 1$ iff $k[g] = 0$, and $h_k^1 = E - V + h_k^0$.
 - Non-trivial holonomy kills H^0 and lowers H^1 by one.
 - The twisted H^0 is exactly the global-section test. The torsion becomes visible over $\mathbb{C}$, which needs complex (or rotation) coefficients unless $n = 2$.

Proof. 1. Shifts commute, and $F^*T_hF = \text{diag}(\omega^{kh})$.

2. The abelianisation of $\text{Aff}(\mathbb{Z}/p)$ is $(\mathbb{Z}/p)^\times$; the example is computed.

3. $\ker \delta$ consists of the flat sections of a rank-one local system, which is non-trivial iff some loop has $\omega^{kh} \neq 1$; then use rank-nullity. □

Observation 29.2 (The mod-3 grid is not a loop-holonomy system). *computed* The grid's contexts overlap as $K_{3,3}$, which has $\beta_1 = 4$. With overlap-conditional transports, every loop operator is the rank-one projection to uniform, with spectrum $\{1, 0, \dots\}$. The spectra are identical for the consistent grid (81 global sections) and the obstructed one (0). The obstruction $\sum \text{rows} \neq \sum \text{cols}$ is one global parity certified by all six contexts together. No proper cycle carries it, so a representation with per-plaquette phases does not exist for this model. The right tools are γ and $\mathbb{F}_p$ elimination.

Proposition 29.3 (The local shadow: certificate phase). *[S]proved; computed* Let every context C carry only local linear data over $\mathbb{F}_p$: a row A_C and a right-hand side b_C . Examples are an operation edge $x_v - ax_u = g$, a parity row $\sum x = r$, and a literal. No global law, section or trivialisation is used.

1. Fredholm alternative: the family glues iff $y \cdot b = 0$ for every certificate y with $yA = 0$. The obstruction is therefore the vector $\Phi(b) = Yb \bmod p$ over a basis Y of certificates. It is a sum of local contributions $y_C b_C$.
2. Local perturbation: $b_C \mapsto b_C + \varepsilon$ changes Φ by $y_C \varepsilon$. Contexts outside $\text{supp}(y)$ do not enter. With a real lift of the local data, the shadow $e^{2\pi i y \cdot b/p}$ is continuous in every b_C .
3. Shifts: the certificates span the cycle space, and $\Phi \neq 0$ iff the holonomy is non-trivial. The loop shadow of Proposition 29.1 is the special case.
4. Grid: there is a single certificate, $y = (-1, -1, -1, 1, 1, 1)$ over rows and columns, so $\Phi = \sum \text{cols} - \sum \text{rows}$. Its support is all six contexts, and every proper sub-family is consistent for every local data.

Proof. Part 1 is linear algebra over a field: $b \in \text{col}(A)$ iff $b \perp \ker A^\top$. Part 2 is linearity. For part 3, the rows $x_v - ax_u$ form the incidence matrix, whose left kernel is the cycle space. Part 4 is computed; removing any context gives full row rank. □

Consequence. The perturbation and every contribution are local. The *detection* of an obstruction, however, needs the whole support of a minimal certificate. That support is a cycle for loop systems and the entire cover for the grid. No statistic computed on a proper sub-family can see the grid obstruction. This is the precise sense in which a local shadow can be complete. Tests are in `tests/test_monodromy.py`.

Theorem 29.4 (Covariant perturbation split). *[S]proved (1–4); 5 as stated Let Γ be connected, with covariant edge channels $K_e = (1 - \eta_e)S_{g_e} + \eta_e U$ on a finite abelian group A . Here g_e is the phase, $\eta_e \in [0, 1]$ is the noise (the magnitude), and U is uniform.*

1. Factorisation. *The loop channel is $K_L = (1 - \eta_L)S_{h_L} + \eta_L U$, with $1 - \eta_L = \prod_{e \in L} (1 - \eta_e)$ and $h_L = \sum_{e \in L} \pm g_e$. On every non-trivial character, $\lambda_\chi(L) = (1 - \eta_L) \chi(h_L)$; on the trivial character it is 1.*
2. Layer 2 reads magnitude only. *Every per-context Shannon or Tsallis quantity, per edge or of the loop channel, is a function of the η_e alone.*
3. The class reads phase only. *The operation class $[g]$, and with it the certificate phase Φ , depends on the g_e alone. The empirical invariants do not: the support of K_e is full whenever $\eta_e > 0$ and is the graph of S_{g_e} when $\eta_e = 0$. So the stratum depends on the magnitudes. $\gamma \neq 0$ requires a non-flat cycle all of whose edges are noiseless; with any noise on every non-flat cycle, $\gamma = 0$ and the model is in the interior.*
4. Split. *Changing the phase of an edge e_0 changes $[g]$ iff e_0 is not a bridge, and changes no Layer-2 quantity. Changing its noise changes Layer 2 and never changes $[g]$.*
5. Coupling through CF (single cycle):
 - (a) *Equal noise, non-flat: $\text{CF} = \max(0, 1 - L\eta/2)$ (Theorem 17.4).*
 - (b) *$A = \mathbb{Z}/2$, per-edge noise: $\text{CF} = \max(0, 1 - \sum_e \eta_e/2)$ if non-flat, and $\text{CF} = \max(0, \frac{1}{2}(\eta_{\max} - \sum_{e \neq e_{\max}} \eta_e))$ if flat. This is Proposition 20.2 in noise coordinates, so a flat cycle is contextual when one edge is noisier than all the others together.*
 - (c) *$A = \mathbb{Z}/p$, $p \geq 3$, per-edge noise: neither formula holds. With one fully uniform edge and the rest noiseless, $\text{CF} = 1 - 1/p$ whether the cycle is flat or not.*

Proof. 1. Use $S_g U = U S_g = U^2 = U$; the product of two covariant channels is covariant with $(1 - \eta_1)(1 - \eta_2)$ and $g_1 + g_2$. Reversed edges are Bayesian inverses, giving S_{-g} . Characters diagonalise shifts, and U kills every non-trivial character.

2. Under uniform input the joint law of an edge is $|A|^{-1} K_e$, whose multiset of entries depends only on η_e . Relabelling invariance (Proposition 16.2) finishes it, and the loop channel has the same form by part 1.
3. $[g]$ is a function of g . The support statements follow from $\eta_e > 0 \Rightarrow K_e > 0$ entrywise, and an edge with full support imposes no constraint. The strata proposition then gives the rest.
4. Changing g_{e_0} by δ changes $[g]$ by δ times the class of e_0 , which is non-zero iff e_0 lies on a cycle.
5. (a) is Theorem 17.4. (b) is Proposition 20.2 with $|s_e| = 1 - \eta_e$; it is computed exact on cycles of length 3–5, with 10 random draws each. (c) is computed. When $\eta_e = 1$ the edge's law is a product, so its non-contextual share is at most the $1/p$ of mass on the consistent diagonal. $\square$

Correction. The per-edge formula $1 - \sum \eta_e/2$ that we had conjectured matched one $\mathbb{Z}/3$ configuration, but it is false for $p \geq 3$. The claim “CF = 0 when $\Phi = 0$ ” is false even for $\mathbb{Z}/2$. The split is clean for Layer 2 against the class. CF couples magnitude and phase *non-linearly*, and depends on both even for flat systems. Tests are in `tests/test_perturbation_split.py`.

Corrections to the proposal we were given:

- $H^1(\Lambda; \mathbb{Z})$ of a space is torsion-free ($\text{Hom}(H_1, \mathbb{Z})$). The torsion in the grid lives in the AMB relative class with coefficients in the $\mathbb{Z}$ -linearised support presheaf.
- A twist by a real 1-cochain squares to zero only when it is closed, and phases $e^{2\pi i/3}$ need $\mathbb{C}$, not $\mathbb{R}$.
- Twisting lowers H^1 rather than making it nonzero.
- For shifts, the monodromy has no permutation part.
- Transports by conditional expectation and their loop spectra are already in Section 15 and the lossy-operations section.

The new content is part 2 of Proposition 29.1, the twisted-Betti identity, and Observation 29.2.

Experiment: **arity-graded monodromy with three sources.** [S]_{measured}
(`examples/vbranch/vmono.py`) **Setup.**

- Level-1 chunks of Corpus V carry a true symbol and an arity k .
- Three sources A, B, C label chunks in their own alphabets: $\pi_s(b)$ with probability $1 - \varepsilon$, uniform otherwise.
- Each edge of the cycle $A-B-C-A$ is compared on a *disjoint* third of 3000 documents. No chunk is labelled by all three, so there is no shared global section.
- *Positive*: on documents shared with A , source C applies a cyclic shift τ of its alphabet to odd-arity chunks only. *Null*: no twist.
- *Estimator*: each edge alignment is a one-to-one assignment on co-label counts (Hungarian), and the holonomy is their composition. It is read on labels whose grade count exceeds twice the estimated noise floor. The arity grades are pooled, k even or odd, and each single k .

grade	null, $\varepsilon = .1$	twist, $\varepsilon = .1$	null, $\varepsilon = .3$	twist, $\varepsilon = .3$
pooled	0.00	0.85 (3.7/8 moved)	0.00	1.00 (4.4/8 moved)
k even	0.00	0.00	0.00	0.00
k odd	0.00	1.00 (7/7 moved)	0.00	1.00 (7/7)
$k = 1, 3, 5$	0.00	1.00 (all moved)	0.00	1.00 (all moved)
$k = 0, 2, 4$	0.00	0.00	0.00	0.00

Table 18: Rate at which the estimated holonomy differs from the identity, over 40 replicates.

Findings.

- In the odd grade the estimated holonomy *equals* the planted twist, conjugated into A ’s alphabet, on every active label (20 of 20 checks).
- *Pooling across arity destroys the phase.* The pooled holonomy is a garbled permutation that moves only half the labels, and the twist appears instead as lost magnitude: $|\lambda_2|$ of the loop

operator falls from 0.573 to 0.425 at $\varepsilon = 0.1$, and from 0.148 to 0.101 at $\varepsilon = 0.3$. This is Theorem 29.4 run in reverse: mixing phases turns into magnitude.

- The arity grading is therefore not decoration. It is the index that keeps the phase readable. A graded twist is invisible as a class to the pooled estimator, and exact to the graded one.

Development note. Two readouts produced false positives and were corrected before the table above. Reading the argmax of the composed loop matrix is biased toward labels that are common in the grade. Symbols with no production of arity k pick up noise-only rows. The table uses the corrected estimator.

Arity grading. In Corpus V, a single parse’s cover (each production instance is a context) is *acyclic*: GYO reduces all 50 sampled parses. So π_1 of the cover is trivial, and an arity-graded monodromy of one tree-generated source is trivial (T0 and Vorob’ev). A non-trivial graded representation needs a cyclic cover, and the three-source experiment above supplies one.

30 Detection: a map of the ideas and theorems

One idea. A contradiction is a failure to glue locally consistent data. The meter keeps three failures apart:

- *self-contradiction*: a unit’s support is empty;
- *overlap disagreement*: $d^0 \neq 0$, the deficit;
- *contextuality*: $d^0 = 0$ but $H^1 \neq 0$, which is “locally consistent, globally inconsistent” [25].

Proposition 30.1 (Localisation). *[S]proved Let g be an operation system with values in an abelian group A .*

1. *On a cycle graph with holonomy $h \neq 0$, for every edge e there is a flat system that differs from g on e alone, namely $g - h \mathbf{1}_e$ with the orientation of e along the cycle. So the edge data do not determine which edge is wrong. On a general graph, the minimal corrections to a flat system are likewise not unique.*
2. *Declaring a spanning tree T trusted removes the ambiguity. There is exactly one flat system that agrees with g on T : its value on each cotree edge is the integral of g along the tree path. The discrepancy sits on the cotree edges of the non-flat fundamental cycles.*

Proof. 1. Changing one edge of the cycle by $-h$ changes the holonomy by $-h$.

2. A flat system is δc for a potential c , and $\delta c = g$ on T determines c up to a constant.

□

Design rules.

1. Build the cover from provenance, not from token distance.
2. Record overlap disagreements; do not average them.
3. Report cycles. Blame an edge only against a declared trusted set (Proposition 30.1).
4. Read phase with loop quantities (holonomy, parity product, deficit), and magnitude with per-context information.
5. Give every detector a planted positive and a null.

Part IV

Status

31 Status of every claim

- **Proved:**

- T0 and its corollary; holonomy obstruction and connecting class;
- Propositions 8–10 of [O]; non-unification; vanishing (trivial action); Corollary 28;
- the p -adic proposition;
- identifiability (Prop. 15.1) and its T0 corollary; the Chernoff bound;
- Theorem 16.1 and Layer-2 blindness; character eigenvalues;
- Theorem 17.4; the lower bound of Theorem 17.5;
- orientation on $\mathbb{Z}/3$ and $\mathbb{Z}/5$; convexity of KL balls and exact pair radii;
- $\text{Aff}(\mathbb{Z}/p)$ invariants; the codimension formula;
- R and E ; the zero set of the deficit;
- support stratification (Proposition 19.1); localisation (Proposition 30.1), its corollary, and two-outcome supports (Proposition 19.3);
- the logical Bell bound (Proposition 20.1); the lower bound in Proposition 20.2; the identities in Remark 20.3; dimension as a cocycle on affine supports over $\mathbb{F}_p$ (Proposition 20.4);
- O2 for the true grammar: Propositions 25.1, 25.2, 25.5, Theorem 25.3, and the root collapse.

- **Computed:**

- V0–V4a identities; H^0 , H^1 , H^2 of small semilattices;
- the mod-3 grid over five rings; possibilistic fragility; Hardy false negative;
- Baudot–Bennequin identities; the Tsallis δ_α cocycle;
- Counterexamples 17.7 and 25.4; the O2 rank table and root collapse.

- **Measured:**

- transformer negative; split–merge recovery; instrument calibrations;
- Wald calibration and N_{80} ; equality in Theorem 17.5;
- wall absorption; $\text{Aff}(\mathbb{Z}/5)$ recovery; graded-extension uniqueness;
- the gated-test table; deficit behaviour; ridge; threshold windows;
- grammar-guided transformers on Corpus B (Section 26);
- equality in Proposition 20.2 for $n = 4$; the dimension deficit from estimated supports; S_2 versus Shannon estimation (no gain); the contradiction-meter benchmark (Section 21); recursive-training collapse (Section 27); unknown branching with chunks and split–merge (Section 28); monodromy and the local certificate phase (Section 29, Proposition 29.3); the covariant perturbation split (Theorem 29.4).

- **Open:**

- O1 identifiability threshold $\theta(V, L, \ell, m)$; O2 on fitted tables (statistical equality of upward rows); O4 power of β_1 ; O6 split-merge at $V = 1024, L = 16$; O7 recurrent-depth transformers;
- Problems 21, 22 (23 blocked; 24 settled);
- $H^{\geq 2}$ for δ and δ_α ;
- the packing proof for Theorem 17.5; an invariant separating noise from holonomy under non-covariant noise;
- integer-coefficient persistence for decorated barcodes;
- the conjecture that operations visible to Layer 2 are lossy and admit no *cohomological* invariant. It is refuted in part: the loop spectrum is a non-cohomological invariant;
- extending the gated test to triple contexts;
- equality in Proposition 20.2 for general n ; the $\mathbb{Z}/n$ modulus-phase analogue; non-triviality of the dimension class; the deficit under non-covariant noise; an operation-twisted Vigneaux complex.

- **Withdrawn or corrected:**

- the embedding probe $[H]$;
- “ $I_2 = \delta H$ ”;
- the two-point entropy formula for noisy shifts;
- “information bounds dictate CF”;
- “the spectrum reads h ”: it is the labelled eigenvalue;
- “the optimal non-contextual part is generally not unique”: it depends on the scenario;
- “deficit independent of CF”: it was then established by the sweep;
- a claim in the nerve module that non-identity bars die only at non-flat events: it fails for $\mathbb{Z}/2$ coefficients and non-abelian groups;
- the “BBV” expansion as a Bell-nonlocality framework: no such framework could be identified;
- the “sheaf-geometric bridge” $\chi(\gamma_{\text{AMB}}) = \text{Hol}(\alpha, \partial M)$ and contextuality at single-outcome supports (Remark in Section 19).

32 Method notes

- Every detector needs a planted positive and a null. The engine’s T0 bug and the thresholded- γ false positives were caught this way.
- Use Wasserstein, not bottleneck, distance for near-degenerate diagrams $[H]$.
- Plug-in MI bias is $(K - 1)(L - 1)/2N$; bootstrap nulls cancel it $[H]$.
- Read the proposition, not the summary.

33 Reproduction

- [H]: the command appendix of the hierarchy paper (`build_corpus_syn.py`, `splitmerge.py`, `estimator4.py`, `estimator6.py`).
- [O]: `holonomy.py`, `amb_gamma.py`, `parity_glue.py`, `h2_rational.py`, `h2_integral.py`.
- [S]: `python -m pytest (143 tests), examples/holonomy_estimated.py, examples/channels.py, examples/o2_ranks.py, examples/o2_verify.py, examples/train_tf.py` with `grid.sh` (JAX), and modules `outcome`, `holonomy`, `channels`, `nerve`, `functors`, `na_contexts`.

References

- [1] S. Abramsky, A. Brandenburger. The sheaf-theoretic structure of non-locality and contextuality. *New J. Phys.* 13:113036, 2011.
- [2] S. Abramsky, S. Mansfield, R. S. Barbosa. The cohomology of non-locality and contextuality. *EPTCS* 95:1–14, 2012.
- [3] S. Abramsky, R. S. Barbosa, K. Kishida, R. Lal, S. Mansfield. Contextuality, cohomology and paradox. *CSL 2015*, LIPIcs 41.
- [4] S. Abramsky, R. S. Barbosa, S. Mansfield. Contextual fraction as a measure of contextuality. *Phys. Rev. Lett.* 119:050504, 2017.
- [5] P. Baudot, D. Bennequin. The homological nature of entropy. *Entropy* 17(5):3253–3318, 2015.
- [6] J. P. Vigneaux. *Topology of Statistical Systems*. PhD thesis, Université Paris Diderot, 2019.
- [7] N. N. Vorob’ev. Consistent families of measures and their extensions. *Theory Probab. Appl.* 7(2):147–163, 1962.
- [8] C. Beeri, R. Fagin, D. Maier, M. Yannakakis. On the desirability of acyclic database schemes. *J. ACM* 30(3):479–513, 1983.
- [9] F. Cagnetta, L. Petrini, U. M. Tomasini, A. Favero, M. Wyart. How deep neural networks learn compositional data: the Random Hierarchy Model. *Phys. Rev. X* 14:031001, 2024.
- [10] A. P. Dempster, N. M. Laird, D. B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *JRSS B* 39(1):1–38, 1977.
- [11] C. F. J. Wu. On the convergence properties of the EM algorithm. *Ann. Statist.* 11(1):95–103, 1983.
- [12] J. K. Baker. Trainable grammars for speech recognition. 1979. K. Lari, S. J. Young. *Comp. Speech Lang.* 4(1):35–56, 1990.
- [13] A. Simpson. Equivalence and conditional independence in atomic sheaf logic. arXiv:2405.11073, 2024.
- [14] T. Fritz. A synthetic approach to Markov kernels, conditional independence and theorems on sufficient statistics. *Adv. Math.* 370:107239, 2020.
- [15] G. Wright. An algebraic perspective on computing with data. PhD thesis, 2023.
- [16] A. Banerjee, S. Merugu, I. S. Dhillon, J. Ghosh. Clustering with Bregman divergences. *JMLR* 6:1705–1749, 2005.
- [17] J. G. Kemeny, J. L. Snell. *Finite Markov Chains*. Van Nostrand, 1960.

- [18] N. Slonim, N. Tishby. Agglomerative information bottleneck. NIPS 1999.
- [19] S. Petrov, L. Barrett, R. Thibaux, D. Klein. Learning accurate, compact, and interpretable tree annotation. ACL 2006.
- [20] H. Edelsbrunner, H. Wagner. Topological data analysis with Bregman divergences. SoCG 2017.
- [21] A. Ostrowski. Über einige Lösungen der Funktionalgleichung $\varphi(x)\varphi(y) = \varphi(xy)$. *Acta Math.* 41:271–284, 1916.
- [22] G. A. Miller. Note on the bias of information estimates. 1955.
- [23] N. Brunner, D. Cavalcanti, S. Pironio, V. Scarani, S. Wehner. Bell nonlocality. *Rev. Mod. Phys.* 86:419, 2014.
- [24] S. L. Braunstein, C. M. Caves. Wringing out better Bell inequalities. *Ann. Phys.* 202:22–56, 1990.
- [25] S. Abramsky. Contextuality: at the borders of paradox. arXiv:2011.04899, 2020.
- [26] L. Fan, M. White, E. Sharma, R. Su, P. K. Choubey, R. Huang, L. Wang. In plain sight: media bias through the lens of factual reporting. *EMNLP 2019*.
- [27] I. Csiszár. I -divergence geometry of probability distributions and minimization problems. *Ann. Probab.* 3(1):146–158, 1975.

role	result	where
<i>What to detect</i>		
obstruction	AMB γ , ring-dependent; sufficient but not necessary (Hardy false negative)	§14
degree	contextual fraction (LP; the dual is a Bell inequality); CF = 1 iff strong	§14, §19
where γ can be nonzero	support stratification; interior versus vertices	Prop. 19.1
binary pairwise claims	all-or-nothing; odd $\mathbb{Z}/2$ cycle = Liar = PR box	Prop. 19.3
operations to outcomes	holonomy $[g]$ and the connecting class	Thm. 16.1
cycles are necessary	acyclic and pairwise consistent implies global (BFMY; Vorob'ev)	Thm. 10.4, §21
<i>What cannot be detected</i>		
information lens	Layer-2 blindness to $[g]$	Prop. 16.2
what it does read	moduli $ s_i $; CF from moduli plus one bit	Prop. 20.2
falsehood	coherent fabrications have CF = 0	§21
which edge	localisation needs a trusted tree	Prop. 30.1
long loops	girth bound $\ell_{\min}$	Thm. 17.5
<i>How to compute</i>		
cheap CF bound	logical Bell bound $\sum p_i - (n - 1)$	Prop. 20.1
noisy cycles	closed form $1 - L \varepsilon/2$	Thm. 17.4
parity claims	Gaussian elimination over $\mathbb{F}_2$ with a certificate	§21
affine operations	dimension deficit $\text{rank}(M - I)$; section count	§20
dimension as a class	cocycle on affine supports over $\mathbb{F}_p$	Prop. 20.4
shortest contradictions	BFS in the parity double cover	§21
<i>From samples</i>		
holonomy	Wald test with delta-method standard error; N_{80}	§15
overlaps	deficit with a pinned bootstrap null; threshold sweep with a null band	§24
meter	pooled flip rate by EM; Bonferroni over cycles	§21
real data	permutation null (BASIL)	§21

Table 19: The detection toolkit, with pointers to where each result is stated.